SCIENCE STUDIO

GRADE 5 • ECOSYSTEMS & MATTER CYCLING



About This Book

A teacher companion for Grade 5 science

PURPOSE

Science Studio brings lesson preparation, teaching moves, and evidence of student thinking together in one place. Use it alongside the official OpenSciEd Teacher Guide, slides, and student materials. The binder helps you find what you need at a glance so you can spend more time listening to students and following their ideas.

BEFORE CLASS

Open the lesson bookmark. Start at the Prep Desk to gather materials, check copy quantities, and open the official slides and media. Read the Science Briefing for the lesson question, the science behind it, and its place in the unit storyline.

WHILE TEACHING

Follow the Teaching Moves in order. Give students time to observe, compare evidence, and explain their ideas. Use the prompts and evidence checks to decide what to ask next. Copy student materials separately using the print directions for each lesson.

BETWEEN CLASSES

Use Pause / Resume to choose a stopping point that fits your schedule. Keep a Field Notes record for each section. Record the last activity completed, what to save, and the first move for next time. Use the reproducible Teaching Notes page for additional space.

HOW THIS BOOK WAS MADE

This companion draws on the OpenSciEd Grade 5 Unit 5.1 Teacher Guide and Student Bundle, Version 1.0. OpenSciEd provides the lesson sequence, investigations, and assessments. Science Studio adds preparation checklists, facilitation cues, and restart notes. The cover and divider artwork was generated for this binder. Use the official photos, diagrams, and data as lesson evidence.

ACKNOWLEDGMENTS

Thank you to the classroom teacher who authored Science Studio for turning everyday teaching decisions into a practical companion. A special thank-you to the fifth-grade team at the Children’s School of Rochester for the care and collaboration you bring to teaching.

How to Use This Book

Follow the same visual cues on the lesson pages. Here’s what each one tells you to do.

01 / HOST SEGMENT

THE SCIENCE BRIEFING

Read before class. Look for the science question, what students already know, what evidence they’ll see, and what should stay open for them to figure out.

02 / PREP DESK

SET UP REQUIRED

Gather materials, check print quantities, and queue the official slides or media.

03 / TEACHING MOVES

Start with the slide and time label. Look below it for the action and, where needed, what to ask and how to respond.

A–B / 5 MIN

Reopen plant-weight patterns.

Example from Lesson 5

04 / PAUSE / RESUME

STOP **SAVE** **RESTART**

Check where to stop, what evidence to keep, and the first move when class resumes.

05 / FIELD NOTES

Use one record per class section. Note the last activity, what students said or showed, and the first question or action for next time.

FIND YOUR PLACE

Use the PDF bookmarks or the printed binder page numbers. Publisher page references name the Teacher Guide or Student Bundle separately.

SCIENCE STUDIO

FALLEN TREE. NEW LIFE.

GRADE 5 / UNIT 5.1 / LESSON 01 / BILL NYE



Setup checklist

SET UP REQUIRED

DO BEFORE CLASS

• Open the Lesson 1 deck and queue the redwood clip. Make four chart spaces for each class. For 22 students, prepare six large sheets for Connect and about 8–11 Investigation Ideas sheets for pairs or trios.

PUBLISHER MATERIALS

Per student:

• Explore a New Phenomenon • Initial Model • Related Phenomena • sticky note • marker

Per small group:

• large (at least 11x17 inches) piece of paper • markers • Investigation Ideas

Per class:

• chart paper • chart paper marker • chart paper markers • Notice and Wonder chart • Initial Consensus Model • Related Phenomena chart

Source: Teacher Guide printed p. 52 (PDF viewer p. 58).

Materials checklist

Use with Prep Desk p. 005 for exact quantities, copy ranges and safety notes.

Class or section: _____ Date: _____

- [ ] Open the official slides and queue the redwood clip; verify playback before class.
- [ ] Set out Notice and Wonder, Initial Consensus Model, Related Phenomena, and Driving Question Board spaces.
- [ ] Prepare individual pages, one Investigation Ideas sheet per pair or trio, and large paper for Connect groups.
- [ ] Gather sticky notes, markers, and student notebooks.

SET ASIDE FOR THE NEXT SECTION

THE SCIENCE BRIEFING

A fallen tree can become a neighborhood. How?

HOST SEGMENT / UNIT 5.1 / LESSON 01

WHERE WE ARE IN THE STORY

We're starting with a fallen tree that looks anything but empty. Give students time to notice what's living there and sketch how it might be connected to the log. Their first models and questions give us a reason to zoom in next time.

Here is the mystery. Open the unit by letting students meet the phenomenon: a fallen tree with other things growing on it. They need time to notice what is visible and wonder about what they cannot see.

1. Make observations | 30 min. Show the photos and redwood clip. Collect observations and questions before introducing the term nurse log.

2. Develop a model | 30 min. Have students sketch how a fallen tree might support other organisms. Compare models and mark ideas that need evidence.

3. Identify related phenomena | 20 min. Invite examples from students' communities. Ask what makes each example similar to the nurse log.

4. Ask questions | 30 min. Cluster student questions on the Driving Question Board. Have students propose ways to investigate them.

Where it leads. Save the models and questions for Lesson 2. The class will take a closer look and start naming the parts of this system.

OPENSCIED UNIT 5.1 / TEACHER GUIDE PRINTED PP. 50–52

Stop and restart

BLOCK 1

Finish Explore. Begin individual models if time allows. Save Notice and Wonder and each student model. Restart at the unfinished M or N step, or at O when models are ready.

BLOCK 2

Finish Synthesize and work through Connect as time allows. Save the consensus model and related phenomena. Finish Connect next time if needed, then open U.

BLOCK 3

Finish Navigate. Save each class Driving Question Board and model. Begin Lesson 2 with one question that closer observation could answer.

PICK UP THE LESSON

Prep Desk / p. 005

Teaching Moves / p. 011

Field Notes / p. 014

Lesson 1 | Print and set up

Print one teacher copy of the companion. For each class, print about 22 sets of individual handout PDF pages 2–5. Print group handout PDF page 2 once per pair or trio for Investigation Ideas. Adjust to actual enrollment. Students can use a notebook T-chart for Notice and Wonder. Open the official Lesson 1 deck. Queue the 19-second Fallen Redwood Tree clip for slide 9. The embedded player appeared blank during preview. The YouTube copy has no captions. Use the visual transcript on Teacher Guide printed page 100 to prepare a description. Check classroom playback. If the clip is unavailable, use the five photos on slide 8 and record observations from the photos. Set up four spaces for each class: Notice and Wonder, Initial Consensus Model, Related Phenomena, and the Driving Question Board. Set out markers, sticky notes, and six large sheets for groups of three or four.

What counts as progress?

Students will see plants and fungi growing on fallen trees. The images let them describe what they see and compare size and location. They cannot yet see what is inside the log or establish how a tree changes into a nurse log. The first model shows students' starting explanations. Keep uncertain ideas visible so Lesson 2 has questions to investigate.

What counts as progress today? Students point to something they observed, show a possible relationship in a model, and name a question a closer look could help answer.

Timing. Explore 30 minutes, Synthesize 30 minutes, Connect 20 minutes, Navigate 30 minutes. Use the publisher's sequence. When a block ends mid-activity, record the last completed slide and save the work named on that activity page.

Explore

SLIDES 2–13 / 30 MINUTES

1. Set up Notice and Wonder

Slide 2

Give students a T-chart. Ask them to write or sketch one thing they see and one thing they wonder. Invite labels in any language.

2. Look closely

Slide 3–7

Display the photographs one at a time. Ask, “Where do you see that?” On slide 6, ask how the people help students judge the size of the log.

3. Compare evidence

Slide 8–9

Use slide 8 to compare the five photos and slide 9 to observe the fallen redwood. Ask students what they noticed on this second look and what remains hard to see.

4. Share and record

Slide 10–12

Have groups rehearse one observation and one question. Record both on the class chart. If a student says, “The log feeds the plant,” ask, “What do you see, and what are you proposing might happen?”

5. Name the phenomenon

Slide 13

Introduce “nurse log” after students have shared their observations. Save the Notice and Wonder chart. Resume with slide 14 when Explore is complete.

Slide 8 photo collection. OpenSciEd Teacher Guide, printed p. 57.

Synthesize

SLIDES 14–19 / 30 MINUTES

1. Explain what a model does

Slide 14

Use the slide’s shadow example. Ask what the model helps explain.

2. Make an individual model

Slide 15

Use both Initial Model pages. Have students show what may happen on, under, and inside the log. Invite arrows and labels. Use question marks for uncertain ideas.

3. Compare with a partner

Slide 16–17

Gather in a Scientists Circle and revisit the class agreements. Have partners explain two similarities and two differences between their models.

4. Build a class model

Slide 18–19

Ask which ideas can go on the model now and which need evidence. If students disagree about whether roots reach through the log, record both possibilities. Ask what a closer observation could reveal.

5. Save the work. Keep the individual models as an ungraded baseline and preserve the class model. Resume at slide 20.

Connect and Navigate

SLIDES 20–25 / 50 MINUTES

minutes Connect · 20 minutes

1. Invite a community example

Slide 20

Ask students to look for something that reminds them of the nurse log. They can talk with a family or community member and bring back an example if able.

2. Compare examples

Slide 21

Give groups large paper. Ask, “What is similar to the nurse log? What is different?” Record the connections students explain. Save the chart and resume at slide 22 if class ends here. Navigate · 30 minutes

1. Write questions

Slide 22

Reopen the charts and model. Have students write one question per sticky note.

2. Build the Driving Question Board

Slide 23

Ask students to read their questions and explain which ones belong together. Keep distinct questions visible within each cluster.

3. Propose an investigation

Slide 24

Have pairs or trios choose a question and suggest something the class could observe. Ask how that observation would help.

4. Choose where to start

Slide 25

Use a student question to motivate taking a closer look at a nurse log in Lesson 2. Save each class’s board and model.

Understanding record

One record per class section

CLASS / SECTION	DATE	ACTIVITY / SLIDE

Source and readiness
Teacher Guide v1.0, printed pp. 50–73 and 100. Official slides A–X are deck positions 2–25. Print estimates use 22 students per class.

Observation students used as evidence

Relationship students proposed in their models

Question a closer look might answer

Last completed activity and saved work

First move next time

SCIENCE STUDIO

INSIDE THE NURSE LOG

GRADE 5 / UNIT 5.1 / LESSON 02 / MR. WIZARD



UNIT SIGNAL / 02 OF 15

Setup checklist

SET UP REQUIRED

DO BEFORE CLASS

• Prepare nurse log, picture and video stations. For six groups, stage two of each type if two terraria are available. Prepare moss, dishes and lenses for eight groups. Read printed pp. 127–128 for rotations, care and cleanup.

PUBLISHER MATERIALS

Per student:

• Explore a Nurse Log • Nurse Log Components in Our Community • My Growing Ideas

Per partners:

• Observe Moss • small piece of moss • hand lens • Observe Moss

Per small group:

• Explore a Nurse Log • hand lens

Per class:

• Explore a Nurse Log • prepared Nurse Log in a Box • Nurse Log Pictures • and Nurse Log Videos stations as described in Nurse Log & Moss Stations Setup • Driving Question Board • initial consensus model from Lesson 1 • “Nurse Log Components in Our Community” chart paper • “Our Questions About Moss” chart paper

Source: Teacher Guide printed p. 104 (PDF viewer p. 110).

Materials checklist

Use with Prep Desk p. 016 for exact quantities, copy ranges and safety notes.

Class or section: _____ Date: _____

- [ ] Prepare the nurse-log observation stations: live log, photos, and videos as available.
- [ ] Stage moss samples, dishes, and hand lenses. Check handling and handwashing guidance.
- [ ] Copy Explore a Nurse Log and other individual pages. Prepare Observe Moss by group or pair.
- [ ] Open the lesson deck and media, and keep each class's station notes and model separate.

SET ASIDE FOR THE NEXT SECTION

THE SCIENCE BRIEFING

Zoom in. What lives in, on, and around the log?

HOST SEGMENT / UNIT 5.1 / LESSON 02

WHERE WE ARE IN THE STORY

Last time we saw a fallen tree supporting other life. Now we're looking closer, using different sources to find what lives in, on, and around the log. The questions students raise about moss will help launch our investigation of plant growth.

Here is the mystery. Use a question from Lesson 1 to give the station work a purpose. Students will compare what different sources let them see.

1. Take stock of our questions | 5 min. Revisit the class questions.

2. Make observations | 20 min. Rotate through the nurse log stations and record what each source shows.

3. Participate in a discussion | 10 min. Add visible components to the model and mark guesses separately.

4. Connect to our communities | 10 min. Connect the observations to things students might see in their community.

5. Plan to explore | 10 min. Turn a wondering about moss into a testable question.

6. Make observations | 10 min. Use hand lenses to compare moss features.

7. Participate in a discussion | 10 min. Ask students to point to the features behind their claims.

8. Create a My Growing Ideas chart | 10 min. Record an idea and its evidence in My Growing Ideas.

9. Revisit our Driving Question Board | 5 min. Return to the Driving Question Board and add questions about plants.

Where it leads. Those questions about moss give Lesson 3 a reason to investigate how plants grow.

OPENSCIED UNIT 5.1 / TEACHER GUIDE PRINTED PP. 101–103

Stop and restart

BLOCK 1

Complete A-F. Save the station observations, model and community chart. Restart at G with what students noticed about moss.

BLOCK 2

Complete G-R. Save observations and testable questions. Open Lesson 3 with a student question about moss or plants.

PICK UP THE LESSON

Prep Desk / p. 016

Teaching Moves / p. 021

Field Notes / p. 023

Print and set up

Lesson 2 | What things are found in, on, and around nurse logs?

PRINT

Print one teacher copy single-sided. Student Bundle PDF viewer pages: Explore a Nurse Log 19-20 (22 sets/class), Observe Moss 21-22 (8 group sets or 11 partner sets/class), My Growing Ideas 23 (22/class), Community Connection 133-134 (22 sets/class if sending home). Adjust to enrollment. Teacher Guide PDF pp. 136-141: two color sets for picture stations.

PREPARE THE LOG STATIONS

Allow the publisher's 60 minutes of preparation. For 22 students, make six groups: four groups of four and two of three. If two terraria are available, set up two live, two photo and two video stations. Start two groups at each type. Rotate in three five-minute rounds, with five minutes for directions and transitions. With one live station, arrange extra time or omit that station as the publisher allows.

PREPARE MOSS

For the publisher's eight-group setup, plan six trios and two pairs. Stage samples, dishes and hand lenses. Organize four rounds so each group sees each moss type. Return moss to its tank and maintain it through at least Lesson 7. Keep Lesson 9 isopods separate from the nurse log kit.

SAFETY AND ACCESS

Follow vendor care sheets and school procedures. Use eyes and hand lenses at the live-log station and keep the lid on between groups. If handling is permitted, review the publisher's precautions on printed p. 109 and its handwashing guidance. Offer sketches, oral descriptions and partner rehearsal.

OPEN BEFORE CLASS

Open the official Lesson 2 deck and queue the four publisher videos linked in Science Studio. Verify playback on the classroom device. Reuse station materials for the second class and keep each class's charts separate.

Block 1

45 minutes | Slides A-F

A / NAVIGATE / 5 MIN

Reopen the Lesson 1 Driving Question Board. Choose an actual class question a closer look could help answer. Ask what students would need to see.

B / EXPLORE / 20 MIN

Rotate through live log, pictures and video. Ask students to label their observation by source and location: in, on or around. Possible observation: Green patches cover part of the wood. Ask where they are and which source shows them.

C-D / SYNTHESIZE / 10 MIN

Add observed components to the initial consensus model. If a student says an animal must live inside, ask what they saw and what would count as evidence. Mark predictions with a question mark. Introduce the system idea by asking how the parts might relate.

E-F / CONNECT / 10 MIN

Invite students to look for familiar components in their community. Save the community chart and model. If class ends here, resume with slide G and the class's questions about moss.

Block 2

45 minutes | Slides G-R

G-J / EXPLORE / 10 MIN

Revisit what students noticed about moss. Help them turn a broad wondering into a question the class can investigate. Ask what they could actually observe.

K-M / EXPLORE / 10 MIN

Invite an initial prediction about moss. Ask partners what feature would help them decide. Compare samples using hand lenses and record visible features on Observe Moss.

N-O / SYNTHESIZE / 10 MIN

Have students point to or sketch the feature behind their claim. If someone says green means it is a plant, ask which other features they can compare. Keep unresolved ideas visible for later evidence.

P / MY GROWING IDEAS / 10 MIN

Have students record one thing they figured out and the observation that supports it. Use drawings, labels or words.

Q-R / NAVIGATE / 5 MIN

Revisit the Driving Question Board. Add new questions about moss and plant growth. Carry those questions into Lesson 3.

Understanding record

One record per class section

CLASS / SECTION	DATE	ACTIVITY / SLIDE

Source and readiness
Teacher Guide v1.0, printed pp. 101–128. Observe Moss discussion and handout are the formative checkpoint. Check deck, classroom playback and kit inventory.

Observations students used as evidence

Testable questions and ideas to revisit

Last completed activity and saved work

First move next time

SCIENCE STUDIO

THE PUMPKIN WEIGH-IN

GRADE 5 / UNIT 5.1 / LESSON 03 / BILL NYE



UNIT SIGNAL / 03 OF 15

Setup checklist

SET UP REQUIRED

DO BEFORE CLASS

• Label three similar-sized, differently shaped mini pumpkins A, B and C. Test the scale and five or six rulers or tapes. Queue the time-lapse and open Scientists Make Arguments.

PUBLISHER MATERIALS

Per student:

• Pumpkin Contest • Investigate How to Fairly Measure Pumpkins • Growing Plants • Observe a Giant Pumpkin Growing • Evaluate the Evidence

Per small group:

• stopwatch or timekeeping device (optional)

Per class:

• Pumpkin Contest • 3 mini pumpkins or other fruit/vegetable • digital scale • 5-6 measuring tapes or rulers • any other available measuring tools • What We Think Plants Need to Grow chart • Giant Pumpkin Timelapse | aka "Bear Swipe" | 2022 All New England Giant Pumpkin Contest Winner video • Classroom Agreements chart from Lesson 1 • Scientists Make Arguments book • Classroom Agreements chart • chart paper or DQB

Source: Teacher Guide printed p. 142 (PDF viewer p. 148).

Materials checklist

Use with Prep Desk p. 025 for exact quantities, copy ranges and safety notes.

Class or section: _____ Date: _____

- [ ] Stage three pumpkins and the measuring tools specified in the Prep Desk.
- [ ] Queue the pumpkin time-lapse and confirm the materials for recording measurements and units.
- [ ] Copy the assigned handouts and prepare Scientists Make Arguments.
- [ ] Set out chart paper and preserve each class’s testable questions.

SET ASIDE FOR THE NEXT SECTION

THE SCIENCE BRIEFING

Big pumpkin! But what does “big” mean?

HOST SEGMENT / UNIT 5.1 / LESSON 03

WHERE WE ARE IN THE STORY

Our close look at moss raised questions about how plants grow. Before we can test what helps them grow, we need a way to measure growth. Use the pumpkins to surface different ideas about measurement and carry a testable approach into the plant investigation.

Here is the mystery. Lesson 2 raised questions about how plants live and grow on nurse logs. Before students can compare growth, they need to agree on what to measure.

1. Recall where we left off | 5 min. Revisit a class question about plants.

2. Connect to our experiences | 15 min. Collect student experiences with growing plants.

3. Make observations | 10 min. Watch the pumpkin time-lapse and separate visible changes from guesses about what caused them.

4. Carry out an investigation | 20 min. Have groups compare three pumpkins with a consistent method and units. Ask what each tool measures if groups get different winners.

5. Read a book | 15 min. Read Scientists Make Arguments and practice asking for evidence.

6. Evaluate our evidence | 15 min. Ask for a measurement and reasoning that support a claim about the fairest measure.

7. Ask questions | 10 min. Save testable questions about plant needs.

Where it leads. Lesson 4 uses the questions to evaluate an investigation and compare actual plant data.

OPENSCIED UNIT 5.1 / TEACHER GUIDE PRINTED PP. 139–141

Stop and restart

BLOCK 1

Pause during I-L after students record their procedure and measurements. Save unfinished comparisons. Restart at the unfinished slide.

BLOCK 2

Finish I-L, then M-R. Save arguments and testable questions to use when evaluating the Lesson 4 setup.

PICK UP THE LESSON

Prep Desk / p. 025

Teaching Moves / p. 030

Field Notes / p. 032

Print and set up

Lesson 3 | How can we measure how much a plant has grown?

PRINT

Print one teacher copy, single-sided. Student Bundle PDF viewer pages: Growing Plants 25-26, Observe a Giant Pumpkin Growing 27-28, Investigate How to Fairly Measure Pumpkins 29-30, and Evaluate the Evidence 31-32. Plan about 22 sets of each selected handout per class. The guide allows notebook entries for the first two.

GATHER

Allow 45 minutes of prep. Label three similarly sized, differently shaped mini pumpkins A, B and C. The guide allows three pieces of the same kind of produce instead. Check the digital scale and five or six rulers or tapes. Stage the scale as a shared station and arrange access for about six groups. Reuse equipment and produce across classes if suitable.

OPEN BEFORE CLASS

Have the official Lesson 3 slides, Pumpkin Contest material, the giant pumpkin time-lapse and Scientists Make Arguments book or projection ready. The exact media and separate resource links still need confirmation. Bring each class's Lesson 2 questions, Classroom Agreements and a What We Think Plants Need to Grow chart.

SCIENCE IN PLAIN LANGUAGE

Length, width and weight can rank the same objects differently. Weight lets students compare the amount of matter in this classroom investigation. Measuring different pumpkins today helps students choose a measure. To find out how much a plant grew, compare measurements of the same plant over time.

Block 1

Slides A-L | First 45 minutes

A / NAVIGATE / 5 MIN

Return to a class question about moss or plant growth. Ask what students would need to observe to show how much a plant has grown.

B-C / CONNECT / 15 MIN

Invite a growing experience. Record ideas about what plants need. If students say water helped, ask what they observed and what they inferred.

D-H / EXPLORE / 10 MIN

Show the publisher's giant pumpkin time-lapse. Have students record visible changes first. Ask which possible growth factors they could see and what the video leaves open.

I-L / EXPLORE / FIRST 15 OF 20 MIN

Ask groups to choose how to judge the biggest pumpkin fairly. Compare all three using the same procedure and units. If a group measures height, ask where they placed the ruler and how another group could repeat the measurement.

AT THE BELL

Save each group's method, data with units, and unfinished comparison. Resume on the last unfinished I-L slide. The suggested pause divides the investigation across blocks and follows the guide's 20-minute total.

Block 2

Finish I-L, then M-R | 45 minutes

I-L / EXPLORE / FINAL 5 MIN

Finish the comparison. If the tallest pumpkin and heaviest pumpkin differ, ask what each tool tells the class. Have students explain which measure best supports a claim about amount of matter.

M-O / CONNECT / 15 MIN

Read Scientists Make Arguments. Use the Classroom Agreements to discuss how students can disagree with a claim and ask for evidence.

P-Q / SYNTHESIZE / 15 MIN

Use Evaluate the Evidence. Ask for a claim, a specific measurement with units, and a reason the measurement supports the claim. For claims about plant needs, ask what comparison would help establish whether a factor affected growth.

R / NAVIGATE / 10 MIN

Add testable questions to the board. Preserve the class's questions for Lesson 4, when students evaluate the plant investigation setup and look at its data.

Understanding record

One record per class section

CLASS / SECTION	DATE	ACTIVITY / SLIDE

Source and readiness
Teacher Guide v1.0, printed pp. 139–144. Student Bundle PDF pp. 25–32. Check classroom materials and playback.

Measurement and reasoning students used

What students inferred from the time-lapse

Testable questions to carry into Lesson 4

Last completed activity and saved work

First move next time

SCIENCE STUDIO

WHAT MAKES PLANTS GROW?

GRADE 5 / UNIT 5.1 / LESSON 04 / MR. WIZARD



UNIT SIGNAL / 04 OF 15

NEXT PAGE / SETUP AND MATERIALS

Setup checklist

SET UP REQUIRED

DO BEFORE CLASS

• Have the Lesson 3 questions ready. Open the investigation setup before displaying plant data or graphs. Prepare the article for the read-aloud.

PUBLISHER MATERIALS

Per student:

• Investigation Setup • Interpret Plant Data • Digging Into Soil • My Growing Ideas (optional)

Per small group:

• Plant Data • Graphs of Plant Data

Per class:

• DQB or testable questions list from Lesson 3 • Plant Growth Investigation

Source: Teacher Guide printed p. 172 (PDF viewer p. 178).

Before class

G172; S33–49, 51. Sort Investigation Setup and Interpret Plant Data for individual use. Prepare Plant Data by pair/trio and Graphs of Plant Data for the gallery if not supplied in the kit. Keep data out of view until after students evaluate the setup. Open Digging Into Soil and retain Lesson 3 questions.

Print and handout map

Student Bundle viewer pages unless marked Teacher Guide printed. Prepare one per student unless otherwise stated (about 22 per class).

Student copies

One Investigation Setup (33–36) for G, Interpret Plant Data (43–46) for L, and Digging Into Soil (47–49) for O per student. My Growing Ideas (51) at P is optional if using notebooks.

Group and teacher copies

One Plant Data set (37–42) per small group. Prepare one Graphs of Plant Data display per assigned gallery station from the kit or publisher resource. Hold the results until students evaluate the setup. Keep Teacher Key (guide printed pp. 198–200 onward) at the teacher station.

Materials checklist

Use with Prep Desk p. 034 for exact quantities, copy ranges and safety notes.

Class or section: _____ Date: _____

- [ ] Queue Plant Growth Investigation and the class questions from Lesson 3.
- [ ] Copy Investigation Setup, Interpret Plant Data, and Digging Into Soil for each student.
- [ ] Sort Plant Data for small groups and prepare the station graphs. Keep results out of sight until needed.
- [ ] Prepare Growing Ideas copies or student notebooks. Keep the teacher key separate.

SET ASIDE FOR THE NEXT SECTION

THE SCIENCE BRIEFING

Which plant comparisons give us a fair test?

HOST SEGMENT / UNIT 5.1 / LESSON 04

WHERE WE ARE IN THE STORY

We've chosen a way to measure growth. Now students have to decide which comparisons actually tell us something about growing conditions. Keep the spotlight on the data and what was held constant. Then leave them with the question that drives Lesson 5: Where did the extra matter come from?

Here is the mystery. Students chose weight as a measure for tracking growth in Lesson 3. Today they use measurements over time and look for comparisons that tell them which conditions mattered.

1. Recall where we left off | 5 min. Return to the questions about plant growth.

2. Evaluate an investigation | 20 min. Examine the investigation setup before showing the results. Ask which plants can be compared fairly.

3. Analyze and interpret data | 35 min. Use the data and graphs to compare changes in weight across conditions. Have students name the specific plants behind their claim.

4. Participate in a discussion | 10 min. Discuss what the comparisons support and where the evidence leaves uncertainty.

5. Read an article | 15 min. Read Digging Into Soil to explore what soil provides.

6. Decide where to go next | 5 min. Ask where the matter came from when plants gained weight.

Where it leads. Lesson 5 investigates whether air, water and light can supply the matter a plant gains.

OPENSCIED UNIT 5.1 / TEACHER GUIDE PRINTED PP. 169–171

Stop and restart

STOP

At a completed J–M comparison, if a pause is needed.

SAVE

Station number, marked graph, and unfinished claim.

RESTART

Reopen that evidence before introducing the soil reading.

Track each section separately

Use the existing understanding record to mark the last activity and saved work. Treat the proposed break as flexible.

PICK UP THE LESSON

Prep Desk / p. 034

Teaching Moves / p. 038

Field Notes / p. 040

Which comparisons make a fair test?

Science and storyline

Lesson 3 established weight as a way to track growth. Compare changes in the same plant over time and use cases with comparable conditions to reason about what helped it grow. The data do not isolate every possible cause.

A–B / 5 MIN

Reopen testable questions.

C–I / 20 MIN

Evaluate conditions before revealing results.

Run it. Show the setup before revealing results. At G, hand out Investigation Setup and have groups record predictions.

Ask. Which conditions differ? What would you keep the same?

Listen and respond. If several variables differ, have students mark both conditions lists and find a more comparable pair. Save the class's fairness criteria.

Teacher Guide printed pages. 177–181

J–M / 35 MIN

Compare weight changes and graph patterns.

Run it. Model the first graph comparison at L before students work through the gallery. Have students record starting and ending weights and conditions on Interpret Plant Data.

Ask. Which two readings support the change?

Listen and respond. If students compare final weights alone, have them mark the starting values. Save a marked graph and unfinished claim if the block ends.

Teacher Guide printed pages. 182–187

Teacher cue. Ask: Which two plants make a fair comparison? Have students name what stayed the same and use starting and ending weights.

Which comparisons make a fair test?

N / 10 MIN

Discuss claims and limits.

Run it. Collect a claim supported by a named comparison. Keep claims visible through the soil reading at O and Growing Ideas at P.

Ask. What can this comparison tell us about soil?

Listen and respond. If students treat healthier growth as proof soil supplied most of the mass, separate the claims. Carry the source-of-matter question to Lesson 5.

Teacher Guide printed pages. 188–193

O–P / 15 MIN

Read about soil.

Teacher cue. Read Digging Into Soil at O. At P, have students record what they figured out and the evidence that supports it, using Growing Ideas or notebooks.

Q / 5 MIN

Save the question of where added matter came from.

Access during instruction

During the data gallery, have partners point to the two compared values and name what stayed the same before writing a claim.

Understanding record

One record per class section

CLASS / SECTION	DATE	ACTIVITY / SLIDE

Likely ideas and next moves
They both grew, so both conditions work equally: ask for starting and ending weights and what else differed. Soil turns directly into most plant matter: ask what the soil reading and the actual comparisons support. Keep the source of added matter open for Lesson 5.

Evidence checkpoint
Ask for the two plants or conditions being compared, the change each showed, and one reason the comparison can or cannot support the claim. If students compare final weights only, ask them to mark the starting values before returning to their claim.

Evidence students used

Ideas or questions to revisit

Last completed activity and saved work

First move next time

SCIENCE STUDIO

MATTER IN THE AIR

GRADE 5 / UNIT 5.1 / LESSON 05 / BILL NYE



UNIT SIGNAL / 05 OF 15

Setup checklist

SET UP REQUIRED

DO BEFORE CLASS

Test the shared digital scale. Confirm it can detect the difference after adding air to the same ball. Use a partially inflated, full-size sports ball that takes a needle, with the approved pump and needle. Follow the guide's equipment directions. Use small plastic syringes for the trapped-air model. Do not substitute balloons or beach balls.

GATHER BY ROLE

For about 22 students, stage roughly six group trays. Each needs an empty paper cup, a cup of water, a flashlight, and a partially inflated sports ball. Provide group access to the shared scale and the Investigating Matter directions. Set out about 11 small plastic syringes for partners and colored pencils or markers for individual models. Prepare the Share Our Findings About Matter and How Can Air Inflate a Basketball? charts. Have the Driving Question Board ready.

PRINT AND REUSE

One Investigate Matter set per student (Student Bundle PDF 53–54) for C–D. One Model to Explain Plants & Matter set per student (PDF 55–56) for L–M. Plan for about 22 of each per class. Bring each student's Interpret Plant Data (PDF 43–46) from Lesson 4 and have the shared results ready for A–B. Reprint only missing copies.

OPEN AND RESET

Open the official Lesson 5 deck and keep Teacher Guide printed pp. 206 and 208 at hand for setup and formative checks. Save the readings and unfinished models for each class, then reset the ball and tray supplies before the next section.

Source: OpenSciEd Teacher Guide printed pp. 206, 208 and 210–223. Printed p. 206 is PDF viewer p. 212.

Materials checklist

Use with Prep Desk p. 042 for exact quantities, copy ranges and safety notes.

Class or section: _____ Date: _____

- [ ] Test the scale and stage the approved pump, needle, and partially inflated sports ball.
- [ ] Set up group trays with a paper cup, a cup of water, flashlight, and ball. Set out small plastic syringes for partners.
- [ ] Copy Investigate Matter and Model to Explain Plants & Matter. Reuse Interpret Plant Data from Lesson 4.
- [ ] Prepare the class data chart, air model chart, and Driving Question Board. Reset trays and the ball between sections.

SET ASIDE FOR THE NEXT SECTION

THE SCIENCE BRIEFING

Where does a plant get the stuff it grows from?

HOST SEGMENT / UNIT 5.1 / LESSON 05

WHERE WE ARE IN THE STORY

The plant data gave us clues about conditions for growth. Today we're asking where the added matter could come from. Let the scale and syringe give students evidence to argue with. Save the question about what light does for the next lesson.

Here is the mystery. A pumpkin gets bigger and heavier. Where did all that extra stuff come from? In Lesson 4, students compared plants grown under different conditions. Those comparisons gave them clues about what helps plants grow. Today they ask a sharper question: Which of those things can supply the matter that becomes part of a plant?

1. Bring back the evidence | 5 min. Reopen the Lesson 4 plant data. Ask which plants gained weight and what students still wonder about where that extra matter came from.

2. Put the candidates to the test | 20 min. Students compare weights with the scale. The basketball is the surprise: ask what changed when they added air. Have groups record actual measurements with units. If the readings barely change, compare measurements and check the setup before drawing a conclusion.

3. Compare the results | 10 min. Pool the class data. Ask, "What result would convince you that air is made of matter?" Water adds measurable weight. Air can too. The flashlight comparison does not show added weight from light. Save the question of what light does for Lesson 6.

4. Make the invisible visible | 25 min. Use the needleless syringe to explore how trapped air takes up space and can be squeezed into a smaller space. Have students sketch particles with less space between them when compressed. Ask where the particles went. If students say they vanished or got smaller, have them compare their model with what the trapped air did.

5. Go back to the pumpkin | 20 min. Have each student model air and water as sources of matter for growth. Ask for an arrow or label that shows where the matter comes from, plus one observation or result that supports the idea.

6. Leave a question for next time | 10 min. Return to the Driving Question Board. Plants need light, too. In Lesson 6, students use solar toys to investigate what light can do and begin to explain how light supplies energy for growth. Later they bring these ideas back to the nurse log model.

At the 45-minute bell. The publisher sequence puts you about 10 minutes into the syringe activity. Pause at a safe point and save the class data chart and any particle sketches. Next time, reopen that work and finish the model. If you already taught part of Lesson 5, resume from the last activity your class actually completed.

SOURCE / OPENSCIED UNIT 5.1 TEACHER GUIDE, PRINTED PP. 203–207



Stop and restart

STOP

Pause at a safe point in the syringe activity.

SAVE

Class chart and any unfinished syringe models.

RESTART

Resume the unfinished F–K activity. For current-year progress, confirm the class and last activity before resuming.

Track each section separately

Use the existing understanding record to mark the last activity and saved work. Treat the proposed break as flexible.

PICK UP THE LESSON

Prep Desk / p. 042

Teaching Moves / p. 046

Field Notes / p. 048

Where does growth get its matter?

Science and storyline

Use Lesson 4 weight comparisons to ask what supplies the extra matter. Air and water are material sources. Light provides energy, which Lesson 6 investigates. A scale reading alone does not prove what light is.

A–B / 5 MIN

Reopen plant-weight patterns.

C–D / 20 MIN

Compare readings and record actual units.

Run it. At C, have partners propose how to weigh water, light, and air. At D, give each student Investigate Matter. In groups of three or four, record predictions, run the comparisons, and revisit predictions in Part C.

Ask. How could we add air to the same ball and measure the change? What should stay the same between readings?

Listen and respond. If students say air cannot be weighed, ask how to trap it. Record the actual readings and units. Use a needle-inflated sports ball, following the guide's equipment directions.

Teacher Guide printed pages. 210–211

E / 10 MIN

Pool results and check variability before explaining.

Run it. Gather students with their handouts around Share Our Findings About Matter. Chart what groups measured separately from their claims.

Ask. Did every group see the same change? What did the scale show?

Listen and respond. If air readings disagree or the change is too small to resolve, check the scale and setup before repeating. Keep the question of what light supplies open.

Teacher Guide printed pages. 212–214

Where does growth get its matter?

F–K / 25 MIN

Model trapped air with a syringe.

Run it. At G, show the plunger with the tip open. At H, have pairs or trios predict and try with the tip covered. Compare observations before revising particle sketches.

Ask. Why can the plunger move partway when air is trapped?

Listen and respond. If students shrink or erase dots, ask where the trapped air could have gone. Compare sketches with the same number of particles. Save unfinished models and the data chart at the bell.

Teacher Guide printed pages. 215–220

Teacher cue. Ask: What changed when you compressed the trapped air? Look for the same number of particles in a smaller space.

L–M / 20 MIN

Model where plant matter comes from.

Run it. Have each student label a source of matter and connect it to the plant on their model.

Ask. Which observation or measurement supports this arrow?

Listen and respond. When students label light as added matter, return to the class chart. Carry the question of what light supplies to Lesson 6.

Teacher Guide printed pages. 221–223

N–O / 10 MIN

Save the question about light.

Access during instruction

During the syringe model, offer dot sketches and oral rehearsal. Ask students to explain what changed and what stayed constant.

Understanding record

One record per class section

CLASS / SECTION	DATE	ACTIVITY / SLIDE

Likely ideas and next moves
Air weighs nothing: compare the same ball before and after adding air. If the scale barely changes, repeat the measurement under consistent conditions and report uncertainty. Particles shrink: ask students to sketch equal numbers of dots in the same trapped sample before and after compression, with different spacing.

Evidence checkpoint
Listen for a measurement supporting the air claim and look for air/water entering the plant model. If an arrow is unlabeled, ask the student to explain what moves and which evidence supports it.

Evidence students used

Ideas or questions to revisit

Last completed activity and saved work

First move next time

SCIENCE STUDIO

POWERED BY LIGHT

GRADE 5 / UNIT 5.1 / LESSON 06 / MR. WIZARD



UNIT SIGNAL / 06 OF 15

Setup checklist

SET UP REQUIRED

DO BEFORE CLASS

• Test the solar toys in the planned sunlight or approved alternate setup. Open the plant-growth video and prepare the modeling handout.

PUBLISHER MATERIALS

Per student:

• Investigate Sunlight Lab • Investigate Sunlight • Observe Plants Growing • Tomatoes for Sale! Buy Fresh Grown Tomatoes Today • Model to Explain Plants & Energy • My Growing Ideas

Per small group:

• Investigate Sunlight Lab • solar toy

Per class:

• Changing the Direction of Light During Plant Growth • Our Ideas About How Light Helps Plants Grow chart paper • What types of changes does sunlight cause? chart paper

Source: Teacher Guide printed p. 228 (PDF viewer p. 234).

Before class

G228 and 248. One tested solar toy per group of 3–4 (about six per section). Try the intended conditions before class. Use the publisher lamp alternative only with school procedures and appropriate equipment. Queue the plant and solar toy media, gather the prior models and set up the discussion chart. See Student Bundle PDF 57, 59, 61, 63 and confirm roles before copying.

Print and handout map

Student Bundle viewer pages unless marked Teacher Guide printed. Prepare one per student unless otherwise stated (about 22 per class).

Student copies

One Observe Plants Growing (57) for the plant video, Investigate Sunlight (59) for the solar toy, and Model to Explain Plants & Energy (61) for the model. Use My Growing Ideas (63) for reflection or continue in notebooks. About 22 sets per section.

Group and teacher prep

One tested solar toy and access to Investigate Sunlight Lab directions per group of 3–4. Queue the direction-of-light media and Tomatoes for Sale reading from the publisher materials. Keep the teacher transcripts and instructional guidance (guide pp. 249–251) handy.

Materials checklist

Use with Prep Desk p. 050 for exact quantities, copy ranges and safety notes.

Class or section: _____ Date: _____

- [ ] Test each solar toy under the planned light conditions. Check the publisher alternative if needed.
- [ ] Queue the plant observation media and the tomato reading.
- [ ] Copy Observe Plants Growing, Investigate Sunlight, and Model to Explain Plants & Energy.
- [ ] Set out the sunlight discussion charts and bring earlier class models.

SET ASIDE FOR THE NEXT SECTION

THE SCIENCE BRIEFING

Light makes things happen. What does it give a plant?

HOST SEGMENT / UNIT 5.1 / LESSON 06

WHERE WE ARE IN THE STORY

Last time students built a case for air and water as sources of matter. Light still needs an explanation. Use the plant observations, solar toys, and reading to distinguish what light does from what air and water provide. At the close, bring that thinking back to the moss questions.

Here is the mystery. Lesson 5 identified air and water as sources of matter for growth. Students now investigate what light contributes.

1. Recall where we left off | 5 min. Discuss what we figured out in the last lesson about how plants get their matter from air and water, but we are unsure how sunlight helps plants to grow.

2. Make observations | 10 min. Watch a video of plants to guide our predictions about how plants use light to grow and discuss our initial ideas about how sunlight can cause changes.

3. Carry out an investigation | 15 min. Observe how the Sun changes solar toys.

4. Participate in a discussion | 15 min. Share our observations and discuss how the energy transferred from sunlight causes changes.

5. Read an article | 20 min. Read to obtain evidence about how plants transfer and use energy from the Sun to grow.

6. Develop a model to explain | 15 min. Individually, we model to explain that a plant (of our choice) transfers sunlight for energy to grow, using air and water for matter.

7. Update our Growing Ideas charts | 5 min. Individually update our Growing Ideas charts to document our new ideas about how plants use matter and energy to grow.

8. Decide where to go next | 5 min. Summarize our ideas about plant growth and think about how what we have figured out about plants connects back to the nurse log.

Where it leads. In Lesson 7, students use their evidence about matter and energy to revise the nurse-log model.

OPENSCIED UNIT 5.1 / TEACHER GUIDE PRINTED PP. 225–227

HOST SEGMENT

Stop and restart

STOP

The planned 45-minute boundary follows F–H.

SAVE

Solar-toy observations and the discussion chart.

RESTART

Open I–K. Ask what the reading can tell us about plants that the toys cannot.

Track each section separately

Use the existing understanding record to mark the last activity and saved work. Treat the proposed break as flexible.

PICK UP THE LESSON

Prep Desk / p. 050

Teaching Moves / p. 054

Field Notes / p. 056

What does light provide to a plant?

Science and storyline

Lesson 5 identified air and water as sources of matter. Sunlight supplies energy. Solar toys provide evidence that light can cause change; the reading supplies the evidence about how plants use that energy.

A / 5 MIN

Reopen the light question.

B–D / 10 MIN

Observe plants responding to light.

Run it. Use Observe Plants Growing to record what changes when light direction changes. Give students time to distinguish the visible change from their explanation.

Ask. What could we observe directly in the video?

Listen and respond. Record claims about light entering as matter as ideas to test. Keep the observations for the next comparison.

Teacher Guide printed pages. 234–235

E / 15 MIN

Investigate solar toys.

Run it. Give each group the solar toy and each student Investigate Sunlight. Compare movement under the assigned light conditions and record the result.

Ask. What changed when the light changed? What stayed the same?

Listen and respond. If the toy does not respond, check the equipment and light condition before interpreting. Keep observations for F–H.

Teacher Guide printed pages. 236–240

What does light provide to a plant?

F–H / 15 MIN

Discuss changes and energy.

I–K / 20 MIN

Read how plants use sunlight.

Run it. Reopen the toy observations. Use the tomato reading to connect energy transfer to plants and ask students to carry that evidence into L–M.

Ask. What can the reading tell us about plants that the toy could not?

Listen and respond. If students label light as plant material, ask what supplies matter and what supplies energy. At O, return to the moss questions.

Teacher Guide printed pages. 242–248

L–M / 15 MIN

Model light, air and water.

N / 5 MIN

Update Growing Ideas.

O / 5 MIN

Close by returning to the nurse log.

Teacher cue. Use students' models and Growing Ideas to review what plants need. Return to the class's moss questions for Lesson 7.

Access during instruction

After the solar-toy trial, offer a two-frame sketch of the same toy in two conditions. Ask students to label what changed and what light supplied.

Understanding record

One record per class section

CLASS / SECTION	DATE	ACTIVITY / SLIDE

Likely ideas and next moves
Light is the plant's food: ask what the toy observation shows and what the reading adds. The toy and plant work in exactly the same way: compare only what the evidence supports, then ask how the reading describes the plant.

Evidence checkpoint
Look for an energy arrow from sunlight and separate matter arrows for air and water. If students label all arrows as matter, ask them to trace what becomes part of the plant and what supplies energy.

Evidence students used

Ideas or questions to revisit

Last completed activity and saved work

First move next time

SCIENCE STUDIO

PLANTS ON THE LOG

GRADE 5 / UNIT 5.1 / LESSON 07 / BILL NYE



Setup checklist

SET UP REQUIRED

DO BEFORE CLASS

• Gather the earlier class models and charts, and prepare the checklist space. Separate the plant assessment and rubric from the regular handouts.

PUBLISHER MATERIALS

Per student:

• My Growing Ideas charts and investigation handouts from prior lessons • Revised Nurse Log Model • What helps plants grow in our community? • Plants Growing in Different Places • Self-Evaluation Rubric • 1-2 sticky notes

Per class:

• charts and models from Lessons 3-6 • Gotta-Have-It Checklist chart paper • Class Consensus Model chart paper • Driving Question Board

Source: Teacher Guide printed p. 255 (PDF viewer p. 261).

Before class

G255; Student Bundle PDF 65–68, 135–136 and 143–146. Bring investigation records, earlier models and Growing Ideas. Prepare checklist/model space and 1–2 sticky notes per student. Separate the individual assessment and rubric. If color copies are unavailable, the guide identifies slide H as an image option.

Print and handout map

Student Bundle viewer pages unless marked Teacher Guide printed. Prepare one per student unless otherwise stated (about 22 per class).

Student practice copies

One Revised Nurse Log Model (65–66) per student for model revision. Bring earlier investigation pages and Growing Ideas. Copy the community page (135–136) per student if assigning it.

Hold for assessment

One Plants Growing in Different Places (143–146) and one Self-Evaluation Rubric (67–68) per student. Distribute the task at G and use the self-check when directed. Slide H can display the photos if color copies are unavailable. Keep the scoring guide (printed pp. 275 onward) teacher-only.

Materials checklist

Use with Prep Desk p. 058 for exact quantities, copy ranges and safety notes.

Class or section: _____ Date: _____

- [ ] Gather earlier investigation pages, Growing Ideas, and the class models.
- [ ] Prepare the class checklist, consensus model space, and sticky notes.
- [ ] Copy Revised Nurse Log Model and any assigned community page.
- [ ] Hold the transfer task and student self-evaluation rubric for the appropriate step. Keep teacher scoring separate.

SET ASIDE FOR THE NEXT SECTION

THE SCIENCE BRIEFING

Bring the evidence back to the log.

HOST SEGMENT / UNIT 5.1 / LESSON 07

WHERE WE ARE IN THE STORY

We've gathered evidence about air, water, and light. Now ask students to put those ideas to work in a model of the nurse log. Their explanations tell us what they can account for about plants. Next we'll shift our attention to the changing log and the animals that live there.

Here is the mystery. Students now have evidence about matter from air and water and energy from sunlight. Have them use those ideas to explain how plants live and grow on a nurse log.

1. Recall where we left off | 5 min. Revisit what we have figured out so far about plants.

2. Build a Gotta-Have-It Checklist | 10 min. Collaboratively build a Gotta-Have-It Checklist to identify what we need to include as we model our new ideas about plants.

3. Develop a model to explain | 25 min. Individually and then model our new ideas about plants in the nurse log system.

4. Connect to our communities | 10 min. With a trusted community member, we identify evidence of objects or factors in our community that help plants grow.

5. Argue from evidence | 30 min. Individually, we write a scientific argument to identify environments that are best and least able to support plant growth.

6. Revisit our Driving Question Board | 10 min. Revisit our DQB to see what questions we have answered and determine where we should go next.

Where it leads. Save the revised models. Lesson 8 shifts attention to changes in the log and the animals in the system.

OPENSCIED UNIT 5.1 / TEACHER GUIDE PRINTED PP. 252–254

Stop and restart

STOP

Aim to finish the model before pausing.

SAVE

Checklist, model revisions, and any notes from F.

RESTART

Resume F if unfinished. Protect the G–I argument and assessment window.

Track each section separately

Use the existing understanding record to mark the last activity and saved work. Treat the proposed break as flexible.

PICK UP THE LESSON

Prep Desk / p. 058

Teaching Moves / p. 062

Field Notes / p. 064

How does a nurse log help plants grow?

Science and storyline

Return to the class model with evidence from Lessons 3–6. A nurse log can provide conditions that support growth. Plants acquire material chiefly from air and water and use energy from light.

A / 5 MIN

Revisit plant findings.

B–C / 10 MIN

Build the checklist.

Run it. Reopen investigation records and Growing Ideas. Build the checklist from findings students can support before they revise models.

Ask. Which observation or comparison belongs behind this checklist idea?

Listen and respond. If a claim is general, have students point to one artifact and name the finding it shows. Keep unresolved ideas visible.

Teacher Guide printed pages. 262–264

D–E / 25 MIN

Revise individual and class models.

Run it. Give students time to revise individual models. Use selected examples in the consensus discussion and ask students to trace their arrows.

Ask. Which part shows matter entering a plant? Which shows the source of energy?

Listen and respond. If an arrow is unlabeled, have its author trace it aloud and add a label. Save a revision to revisit.

Teacher Guide printed pages. 265–268

Teacher cue. Ask students to point to a label or arrow and explain which investigation supports it before they revise.

How does a nurse log help plants grow?

F / 10 MIN

Connect to community evidence.

G–I / 30 MIN

Argue from evidence as directed.

Run it. Distribute Plants Growing in Different Places at G. Preserve individual argument writing and use the publisher self-evaluation rubric when directed.

Ask. Clarify directions within the publisher guidance and save instructional prompts for afterward.

Listen and respond. Use the scoring tool to identify missing evidence connections. Follow up after collecting the individual work.

Teacher Guide printed pages. 271–272

J–K / 10 MIN

Revisit questions.

Access during instruction

During model practice before the transfer task, have partners point to an arrow and rehearse which evidence supports it.

Understanding record

One record per class section

CLASS / SECTION	DATE	ACTIVITY / SLIDE

Likely ideas and next moves
The log provides all the plant's matter: ask students to trace air and water arrows and name evidence for each. A model is better because it has more labels: ask which changed relationship the evidence supports.

Evidence checkpoint
Have a student explain one specific change from the earlier model and the observation behind it. If the explanation lacks evidence, prompt a return to the relevant investigation record before the independent task.

Evidence students used

Ideas or questions to revisit

Last completed activity and saved work

First move next time

SCIENCE STUDIO

THE GHOST LOG

GRADE 5 / UNIT 5.1 / LESSON 08 / MR. WIZARD



Setup checklist

SET UP REQUIRED

DO BEFORE CLASS

• Check the nurse-log terrarium and hand lenses. Pull the earlier moss questions and queue the nurse-log book and termite colony data.

PUBLISHER MATERIALS

Per student:

• Nurse Logs Over Time • Analyze Data of a Termite Colony

Per partners:

• Weight of a Termite Colony in a Closed System

Per small group:

• hand lenses

Per class:

• Driving Question Board • Nurse Log in a Box • Notice and Wonder chart • Our Questions About Moss chart (from Lesson 2) • Testable Questions and Evidence to Collect chart paper (optional) • Nurse Logs: More than Meets the Eye • Extension questions for termite data • Analyze Termite Data: Teacher Key

Source: Teacher Guide printed p. 281 (PDF viewer p. 287).

Before class

G281. Check the Nurse Log in a Box and stage lenses per guide. Pull DQB and earlier moss questions. Have the nurse-log book or projection ready, plus termite data and student recording pages (Student Bundle viewer 69–77, 79). Sort partner data from individual recording pages.

Print and handout map

Student Bundle viewer pages unless marked Teacher Guide printed. Prepare one per student unless otherwise stated (about 22 per class).

Student copies

One Nurse Logs Over Time (69–72) for observation stations and Analyze Data of a Termite Colony (73–77) for O. Copy My Growing Ideas (79) for reflection or use notebooks.

Pair and teacher copies

One Weight of a Termite Colony in a Closed System graph per pair or small group, from the publisher resource, to accompany O. Keep Analyze Termite Data: Teacher Key (printed pp. 307–308) and optional extension (pp. 309–310) separate.

Materials checklist

Use with Prep Desk p. 066 for exact quantities, copy ranges and safety notes.

Class or section: _____ Date: _____

- [ ] Inspect Nurse Log in a Box and stage hand lenses and observation stations.
- [ ] Have the nurse-log reading and earlier moss questions ready.
- [ ] Copy Nurse Logs Over Time and Analyze Data of a Termite Colony for each student.
- [ ] Prepare colony graphs by pair or small group and keep the teacher key separate.

SET ASIDE FOR THE NEXT SECTION

THE SCIENCE BRIEFING

The log looks smaller. Where did its matter go?

HOST SEGMENT / UNIT 5.1 / LESSON 08

WHERE WE ARE IN THE STORY

So far the log has been a place where plants grow. Now look at the log itself: it seems to shrink. Students compare observations and termite data to look for where the matter might go. That sets up the next question about what animals get and use from food.

Here is the mystery. The class has modeled plant growth on the log. Now they investigate why a log seems to disappear and what might happen to its matter.

1. Take stock of our questions | 5 min. Consider what other questions we had about nurse logs, especially how nurse logs change over time.

2. Make observations | 15 min. With the class divided into two halves, we observe the Nurse Log in a Box and a series of images showing nurse logs gradually seeming to disappear.

3. Ask questions | 10 min. Generate new claims and testable questions about the unexplained phenomenon of ghost logs.

4. Read a book | 15 min. Read a book about how nurse logs help animals meet their needs for survival, with a focus on those animals that eat the log itself.

5. Make sense of data | 20 min. Analyze data from a closed system containing termites and a nurse log.

6. Participate in a discussion | 20 min. Share our findings in a discussion to answer the lesson question, “Why does a nurse log seem to lose matter over time?”.

7. Decide where to go next | 5 min. Individually, we begin to consider what else animals do with the food that they eat, other than gain matter.

Where it leads. The termite evidence sets up Lesson 9’s question about what animals do with the energy they get from food.

OPENSCIED UNIT 5.1 / TEACHER GUIDE PRINTED PP. 278–280

Stop and restart

STOP

Finish M before stopping.

SAVE

Observations, book evidence, and competing questions.

RESTART

Begin N–O with the first data comparison. Termite data notes are produced in Block 2.

Track each section separately

Use the existing understanding record to mark the last activity and saved work. Treat the proposed break as flexible.

PICK UP THE LESSON

Prep Desk / p. 066

Teaching Moves / p. 070

Field Notes / p. 072

Where does the matter in a shrinking log go?

Science and storyline

Lesson 7 explained plant growth. Now the log looks smaller over time. Appearance can start a question; termite colony evidence is needed to reason about where matter might move.

A / 5 MIN

Take stock of questions.

B–I / 15 MIN

Observe changes.

Run it. Have students record observations by station on Nurse Logs Over Time. Label the source and separate observed change from explanations.

Ask. Which source shows a change? What could tell us where the matter went?

Listen and respond. If students say matter disappeared, put that idea on the question list for comparison with the book and colony data.

Teacher Guide printed pages. 287–290

J–L / 10 MIN

Ask new questions.

M / 15 MIN

Read the book.

Where does the matter in a shrinking log go?

N–O / 20 MIN

Interpret colony data.

Run it. At O, give each student the analysis handout and share the colony graph by pair or group. Model locating one component and reading the units.

Ask. What happened to the weight of one component between these readings?

Listen and respond. If students read across unlike components, trace one row over time. Record two values before interpreting.

Teacher Guide printed pages. 296–299

Teacher cue. At O, orient students to the colony data. Ask them to cite two readings before making a claim about a weight trend.

P–T / 20 MIN

Discuss how matter moves.

Run it. Compare changes in the log and other components. Let students label a possible destination for matter and identify which part is an inference.

Ask. What did we measure? What destination does the evidence suggest?

Listen and respond. If students expect one gain to equal the entire log loss, name the measured parts and what the data cannot track.

Teacher Guide printed pages. 300–305

U / 5 MIN

Decide what comes next.

Access during instruction

During the termite-data discussion, let students trace a row with a finger or ruler, then say the trend with two readings before writing.

Understanding record

One record per class section

CLASS / SECTION	DATE	ACTIVITY / SLIDE

Likely ideas and next moves
The wood disappeared: ask what the picture lets us see and what it cannot track. The termite weight gain proves where all the wood went: ask what the data track and what remains outside the evidence.

Evidence checkpoint
Ask students to connect one data pattern with a possible transfer of matter. If they use a photo as proof of a destination, return to the measurements and separate the two evidence sources.

Evidence students used

Ideas or questions to revisit

Last completed activity and saved work

First move next time

SCIENCE STUDIO

ISOPODS ON THE MOVE

GRADE 5 / UNIT 5.1 / LESSON 09 / BILL NYE



Setup checklist

SET UP REQUIRED

DO BEFORE CLASS

• Set up the isopod terrarium, lamp and thermometers using the publisher reference. Check school handling and heat-lamp procedures before class.

PUBLISHER MATERIALS

Per student:

• Investigate Isopods Lab • Investigate Isopods handout • Investigate Isopods • Feathered Friends • Feathered Friends Evidence Table

Per class:

• Investigate Isopods Lab • terrarium with isopods • terrarium lid or plastic tub • heat lamp • two thermometers • print out or re-creation of the Temperature for Isopod Investigation reference • stopwatch or other timekeeping device

Source: Teacher Guide printed p. 317 (PDF viewer p. 323).

Before class

G317–319, 338 and following setup pages; Student Bundle PDF 81–83, 85–92. Allow 60–90 minutes prep. Default to the publisher demonstration: isopod terrarium, clear lid or tub, heat lamp, two thermometers, temperature reference and timer. Follow care, school and heat-lamp procedures; reset and check the setup for section two. Publisher video is the approved alternative if the live setup is unavailable. Separate lab pages and bird reading/evidence pages; prepare individual copies as directed.

Print per student

Investigate Isopods, Student Bundle PDF 81–83, before data collection. For N–O, give every student one assigned bird article (PDF 85–86, 87–88, or 89–90) and one Evidence Table (PDF 91–92). Split about 22 article copies among the three birds, based on reading pairs and mixed groups.

Shared demonstration

Print or recreate one Warm/Cool temperature reference for the demonstration (Teacher Guide printed p. 340). Reuse it for both classes. Confirm the video alternative before class if the live demo is unavailable.

Teacher only

Keep the live setup and care reference (printed pp. 338–339) and formative guidance (p. 319) at the demo station. Save class data before handing out the next block's readings.

Materials checklist

Use with Prep Desk p. 074 for exact quantities, copy ranges and safety notes.

Class or section: _____ Date: _____

- [ ] Set up the isopod demonstration with the terrarium, heat lamp, two thermometers, reference, and timer. Check school procedures.
- [ ] Confirm the publisher video option if the live setup is unavailable.
- [ ] Copy Investigate Isopods, one assigned bird reading per student, and the evidence table.
- [ ] Stage the class data record and plan safe closure and reset between sections.

SET ASIDE FOR THE NEXT SECTION

THE SCIENCE BRIEFING

What do animals do with energy from food?

HOST SEGMENT / UNIT 5.1 / LESSON 09

WHERE WE ARE IN THE STORY

The termite evidence helped us think about where matter goes when an animal eats. Today the isopod investigation and bird readings bring energy into the story. Have students keep what they observed separate from what they infer. They'll use both kinds of evidence to explain animal survival next time.

Here is the mystery. Lesson 8 traced matter from wood to termites. Today students compare observations of animal behavior and ask how energy from food helps animals live.

1. Recall where we left off | 10 min. Revisit and share our ideas about what animals do with the food they eat.

2. Plan to explore | 10 min. Share what we know about pillbugs and isopods and learn more key details about this group of animals.

3. Carry out an investigation | 15 min. Investigate how isopods use energy when heat and light are introduced into their habitat.

4. Participate in a discussion | 10 min. Interpret our data from the isopod investigation and connect it to the nurse log habitat.

5. Read an article | 20 min. In pairs, we read and share about how birds keep warm in the winter to highlight other ways that some animals use energy to balance their body temperature.

6. Participate in a discussion | 20 min. Use what we have learned about how isopods and birds get and use energy to figure out what we can now say about animals in general.

7. Take stock of what we figured out | 5 min. Review our new understanding to determine what we can explain about the nurse log in the next lesson.

Where it leads. Keep the animal evidence for Lesson 10, when students use it to explain how the nurse-log system helps animals.

OPENSCIED UNIT 5.1 / TEACHER GUIDE PRINTED PP. 314–316

Stop and restart

STOP

Finish or safely close the live demonstration before the bell.

SAVE

Demonstration data and the first discussion claims.

RESTART

Start N–O with one unresolved question about animal energy.

Track each section separately

Use the existing understanding record to mark the last activity and saved work. Treat the proposed break as flexible.

PICK UP THE LESSON

Prep Desk / p. 074

Teaching Moves / p. 078

Field Notes / p. 080

What does an animal do with energy from food?

Science and storyline

Lesson 8 traced matter from wood to termites. Here students investigate uses of energy and later trace it through food back to the Sun. Movement and temperature observations do not directly measure energy from food.

A / 10 MIN

Revisit animals and food.

B–F / 10 MIN

Introduce isopods and plan observations.

G–J / 15 MIN

Run the demonstration or publisher video.

Run it. Introduce the demo setup and take predictions before showing the result. Have students record temperature and movement observations on Investigate Isopods.

Ask. What change in movement would help us compare predictions?

Listen and respond. Accept contrasting predictions. For the publisher video, pause at evidence points for recording. Follow publisher and school guidance to close the live setup.

Teacher Guide printed pages. 323–327

What does an animal do with energy from food?

K–M / 10 MIN

Discuss temperature and movement evidence.

Run it. Gather readings and behavior observations before explaining the response. Compare the warm condition to the publisher's habitat reference.

Ask. What changed in the setting, and what did the animals do?

Listen and respond. If students say the lamp gave food energy, separate external conditions from energy obtained from food. Save an unresolved question for the bird comparison.

Teacher Guide printed pages. 328–329

Teacher cue. Ask for one temperature reading and one movement observation. Keep observations separate from claims about energy from food.

N–O / 20 MIN

Read the bird article.

Run it. Give each student one assigned bird reading. Have students record evidence and share findings in mixed groups before the class comparison.

Ask. How does this bird use energy? Which detail supports the idea?

Listen and respond. If students generalize from one bird, have them compare another reading before synthesizing.

Teacher Guide printed pages. 330–332

P–R / 20 MIN

Compare isopods and birds.

S / 5 MIN

Take stock.

Access during instruction

After the demonstration, let partners point to a thermometer reading or recorded observation and rehearse a claim that distinguishes temperature from movement.

Understanding record

One record per class section

CLASS / SECTION	DATE	ACTIVITY / SLIDE

Likely ideas and next moves
They moved because they just ate: ask what changed in the demonstration and what was actually measured. The lamp gave them food energy: ask students to trace food energy separately from the warming of the environment.

Evidence checkpoint
At K–M, look for a specific observation or reading supporting the claim. At P–R, ask students to trace food energy back through plants to sunlight. If the chain breaks, have them point to a relevant part of the reading or comparison chart.

Evidence students used

Ideas or questions to revisit

Last completed activity and saved work

First move next time

SCIENCE STUDIO

LIFE AROUND THE LOG

GRADE 5 / UNIT 5.1 / LESSON 10 / MR. WIZARD



UNIT SIGNAL / 10 OF 15

Setup checklist

SET UP REQUIRED

DO BEFORE CLASS

• Gather the animal evidence and the checklist. Set aside the salamander transfer task and its teacher scoring guide for the assessment stop.

PUBLISHER MATERIALS

Per student:

• My Growing Ideas charts from Lesson 8 (and optionally added into Lesson 9) • handouts from previous investigations • Revised Nurse Log Model • Animals in Our Community • Ensatina Salamanders & Nurse Logs • 1-2 sticky notes

Per partners:

• Revised Nurse Log Model

Per small group:

• Lesson 10 Gotta-Have-It Checklist

Per class:

• Driving Question Board (optional) • Class Gotta-Have-It Checklist from Lesson 7 • charts and evidence tables from Lessons 8 and 9 • Nurse Log Consensus Model chart •

DQB

Source: Teacher Guide printed p. 347 (PDF viewer p. 353).

Before class

G347; Student Bundle PDF 93–96, 137–138 and assessment 147–150. Bring models, Growing Ideas and earlier evidence tables. Prepare class checklist and 1–2 sticky notes/student. Print assessment sets separately. Use the publisher scoring and permitted-support guidance; keep it teacher-only.

Print and handout map

Student Bundle viewer pages unless marked Teacher Guide printed. Prepare one per student unless otherwise stated (about 22 per class).

Student and group practice

One Revised Nurse Log Model (95–96) per student. Partners review their individual models. Prepare one Gotta-Have-It Checklist (93–94) per small group for the opening Explore. Copy Animals in Our Community (137–138) per student if assigning it.

Hold for assessment

One Ensatina Salamanders & Nurse Logs (147–150) per student for slide I. Keep it separate from practice and follow the official directions for administration. Keep the scoring guide (printed pp. 364 onward) teacher-only.

Materials checklist

Use with Prep Desk p. 082 for exact quantities, copy ranges and safety notes.

Class or section: _____ Date: _____

- [ ] Gather models, Growing Ideas, and evidence from Lessons 8 and 9.
- [ ] Prepare the Gotta-Have-It Checklist, class model space, and sticky notes.
- [ ] Copy individual Revised Nurse Log Model pages and the checklist for groups.
- [ ] Hold the Ensatina Salamanders & Nurse Logs assessment for slide I and keep scoring guidance teacher-only.

SET ASIDE FOR THE NEXT SECTION

THE SCIENCE BRIEFING

Can your model explain how an animal survives here?

HOST SEGMENT / UNIT 5.1 / LESSON 10

WHERE WE ARE IN THE STORY

We've followed what animals get from food and how they use it. Today students bring that evidence back to the nurse-log model, then apply their reasoning to a new animal. Keep their explanations visible so the termite investigation can pick up the question of how animals use their food.

Here is the mystery. Students bring together their evidence about how animals get and use matter and energy. The model should show how the animal's needs connect to specific parts of the system.

1. Recall where we left off | 5 min. Navigate from the observations and questions we shared in the last lesson to today's work of putting the pieces together for animals in the nurse log system.

2. Build a Gotta-Have-It Checklist | 15 min. In small groups and then collaboratively build a Gotta-Have-It Checklist.

3. Develop a model to explain | 20 min. Individually, we model to explain our ideas about how nurse logs help animals live and grow, then provide feedback to another student on their model.

4. Participate in a discussion | 20 min. Participate in a Consensus Discussion as we model to explain what we agree on and what we still have questions on to explain how nurse logs help animals live and grow.

5. Connect to our communities | 5 min. Outside of class, we look for evidence of animals getting and/or using matter and energy in our community.

6. Apply ideas to another phenomenon | 20 min. Individually, we complete an assessment in which we develop and use a model that describes how animals use matter and energy from food.

7. Revisit our Driving Question Board | 5 min. Revisit our Driving Question Board to see what questions we have answered and determine where we should go next.

Where it leads. The salamander task gives students a chance to transfer what they learned. Lesson 11 follows a new question about how termites can use wood as food.

OPENSCIED UNIT 5.1 / TEACHER GUIDE PRINTED PP. 344–346

Stop and restart

STOP

If F cannot start and finish productively, pause after D–E.

SAVE

Models and feedback.

RESTART

Compare one pair of ideas. Protect the assessment window, allowing extra time if needed.

Track each section separately

Use the existing understanding record to mark the last activity and saved work. Treat the proposed break as flexible.

PICK UP THE LESSON

Prep Desk / p. 082

Teaching Moves / p. 086

Field Notes / p. 088

How does the system help an animal survive?

Science and storyline

Use the termite and isopod/bird evidence to link an animal's needs to parts of the nurse-log system. Use the official salamander task to assess transfer separately.

A / 5 MIN

Reopen animal evidence.

B–C / 15 MIN

Build checklist.

Run it. Use prior models, charts, and evidence tables to build the animal checklist in groups before students draw.

Ask. What does the animal get from its surroundings? What evidence shows that?

Listen and respond. If students list habitat features alone, ask which need each feature meets.

Teacher Guide printed pages. 351–353

D–E / 20 MIN

Model and give peer feedback.

Run it. Have students draft individually. Partners trace a connection and give evidence-based feedback before revision.

Ask. Where does this arrow start and end? Which observation supports it?

Listen and respond. If the arrow only links two organisms, ask the student to label what moves or what resource is provided.

Teacher Guide printed pages. 355–356

Teacher cue. During feedback, ask each pair to identify one connection in the model and name the earlier evidence that supports it.

How does the system help an animal survive?

F / 20 MIN

Discuss models.

G / 5 MIN

Connect to community.

H–I / 20 MIN

Apply ideas to the official new phenomenon.

Run it. Move from class modeling to the new animal. Distribute Ensatina Salamanders & Nurse Logs at I and follow the publisher's independent assessment directions.

Ask. Review each question, invite clarifying questions, and remind students which earlier evidence is available as the guide directs.

Listen and respond. Score the relationship and evidence using the teacher tool. Follow up on incomplete connections after collection.

Teacher Guide printed pages. 361–362

J–K / 5 MIN

Revisit questions.

Access during instruction

During consensus-model practice before the assessment, let students trace a path on the class model and name the evidence for one connection.

Understanding record

One record per class section

CLASS / SECTION	DATE	ACTIVITY / SLIDE

Likely ideas and next moves
The animal gets everything from the log: ask the student to identify one specific need, source and transfer. Peer feedback says add more arrows: ask which relationship an arrow should explain and which evidence supports it.

Evidence checkpoint
Review one model connection before transfer and use the official scoring tool afterward. If the chain lacks a source, point students back to the relevant prior evidence during practice. Preserve the publisher's assessment conditions during H–I.

Evidence students used

Ideas or questions to revisit

Last completed activity and saved work

First move next time

SCIENCE STUDIO

TERMITES AT WORK

GRADE 5 / UNIT 5.1 / LESSON 11 / BILL NYE



Setup checklist

SET UP REQUIRED

DO BEFORE CLASS

• Queue the termite video and prepare Termite Guts for partners. Stage the evidence organizer for the comparison of the video and reading.

PUBLISHER MATERIALS

Per student:

• Termite Eating Wood handout • Evidence for Termite Matter and Energy handout

Per partners:

• Termite Guts handout

Per class:

• Termites Eating Wood video • optional Microscopic View of Termite Gut video

Source: Teacher Guide printed p. 371 (PDF viewer p. 377).

Before class

G371. Queue Termites Eating Wood and the optional microscopic video, verify access, prepare Termite Guts for partners and the evidence organizer. Student Bundle PDF 97–101. Use the guide's notebook option where appropriate.

Print and handout map

Student Bundle viewer pages unless marked Teacher Guide printed. Prepare one per student unless otherwise stated (about 22 per class).

Student and partner copies

One Termite Eating Wood (101) and Evidence for Termite Matter and Energy (99–100) per student. Give the observation sheet with the video and use the organizer across both evidence sources. Prepare one Termite Guts reading (97–98) per pair for the reading activity.

Teacher prep

Queue Termites Eating Wood and the optional microscopic video. Use a notebook in place of a handout only where the guide permits.

Materials checklist

Use with Prep Desk p. 090 for exact quantities, copy ranges and safety notes.

Class or section: _____ Date: _____

- [ ] Queue Termites Eating Wood and the optional microscopic view.
- [ ] Copy Termite Eating Wood and the evidence organizer for each student.
- [ ] Prepare one Termite Guts reading per pair.
- [ ] Set out the evidence organizer or publisher-approved notebook alternative before students compare sources.

SET ASIDE FOR THE NEXT SECTION

THE SCIENCE BRIEFING

How can a termite make a meal out of wood?

HOST SEGMENT / UNIT 5.1 / LESSON 11

WHERE WE ARE IN THE STORY

We've been explaining what animals need to survive. Termites bring us back to a puzzling example: wood is their food. Put the video and reading side by side so students can build an explanation from both. Then ask how another organism on the log, a mushroom, might get what it needs.

Here is the mystery. Students already know animals use food for matter and energy. Wood makes this question interesting. Have them use the video and article to figure out what happens inside the termite.

1. Recall where we left off | 5 min. Discuss what we have figured out about animals, matter, and energy, which makes us wonder how termites can get matter and energy from wood in the nurse log.

2. Make observations | 20 min. Watch a video of a termite eating wood, and then discuss our initial ideas about how the termites use wood as a source of matter and energy.

3. Read an article | 25 min. In partners, we read an article about what happens when termites eat wood to gather evidence to explain how termites get matter and energy from wood.

4. Develop a model | 35 min. Model our ideas based on evidence to explain how termites get matter and energy from wood in the nurse log.

5. Decide where to go next | 5 min. Discuss if other organisms in the nurse log system are able to break down the wood of the nurse log.

Where it leads. Save the explanation about microscopic organisms. In Lesson 12, students turn to other organisms that grow on the log.

OPENSCIED UNIT 5.1 / TEACHER GUIDE PRINTED PP. 368–370

Stop and restart

STOP

If stopping during F–H, note the paragraph reached.

SAVE

Each pair’s evidence table.

RESTART

Begin with a partner recap before finishing the reading.

Track each section separately

Use the existing understanding record to mark the last activity and saved work. Treat the proposed break as flexible.

PICK UP THE LESSON

Prep Desk / p. 090

Teaching Moves / p. 094

Field Notes / p. 096

How do termites use wood as food?

Science and storyline

Build on animals getting matter and energy from food. The video shows termites eating wood. The article supplies evidence about the role of organisms in their gut; a surface video cannot show that mechanism.

A–B / 5 MIN

Revisit animal food evidence.

C–E / 20 MIN

Observe termites and record questions.

Run it. Have students watch Termites Eating Wood and record observations and questions. Revisit selected moments before asking for an explanation.

Ask. What did you see? What can't the video show about what happens inside?

Listen and respond. Move unobserved explanations to questions or inferences and identify what evidence would help.

Teacher Guide printed pages. 375–377

How do termites use wood as food?

F–H / 25 MIN

Read and add evidence.

Run it. Read Termite Guts in pairs and label the source of each detail in the evidence organizer.

Ask. What does the article explain that the video could not?

Listen and respond. If students say wood vanishes, have them locate a detail about what happens to some of the material.

Teacher Guide printed pages. 378–381

Teacher cue. Ask students to mark which evidence came from the reading and which came from the video. Then ask how the two sources fit together.

I–J / 35 MIN

Build and compare models.

Run it. Have students draw the path of material and compare models. Ask them to trace one connection using their organizer.

Ask. Where does the wood start, and where can some of its matter go?

Listen and respond. If they name an outcome without a pathway, connect each arrow to one detail from the reading or video.

Teacher Guide printed pages. 380–385

K / 5 MIN

Identify the next question.

Access during instruction

During the evidence sort, invite students to mark each idea as observed in the video or inferred from the reading, then explain one choice orally.

Understanding record

One record per class section

CLASS / SECTION	DATE	ACTIVITY / SLIDE

Likely ideas and next moves
The termite alone breaks down the wood: ask which part of the article adds a role for the gut organisms. The video proves the full mechanism: ask what is visible and what students learned only from text.

Evidence checkpoint
Ask students to trace wood, microscopic organisms and termite in a model. If a link is unsupported, have them mark the relevant evidence source and clarify what it establishes.

Evidence students used

Ideas or questions to revisit

Last completed activity and saved work

First move next time

SCIENCE STUDIO

MUSHROOM EVIDENCE

GRADE 5 / UNIT 5.1 / LESSON 12 / MR. WIZARD



Setup checklist

SET UP REQUIRED

DO BEFORE CLASS

• Check mushroom kit lead time and safety guidance. Set up the assigned observation sources and keep them labeled for mixed-group comparisons.

PUBLISHER MATERIALS

Per student:

• Investigate Mushrooms • Mushrooms in Our Community

Per small group:

• Investigate Mushrooms • 1-3 mushrooms • hand lenses • mushroom kits • Data and Instructions for Exploring Mushrooms • Investigate Mushrooms

Per class:

• Driving Question Board • Mushroom Comparison chart

Source: Teacher Guide printed p. 394 (PDF viewer p. 400).

Before class

G394, 396, 410 and following setup pages; Student Bundle PDF 103–106 and optional community page

139. Allow 45 minutes prep and a kit grown at least one week, if using a live kit. Prepare three evidence types: close observation (1–3 mushrooms/group), before/after kit or publisher images, and light/dark images. Print one source guide per expert group; individual evidence sheets as directed. For about 22, plan six expert groups, two per source, then balanced mixed groups; assign extra students as co-reporters. Follow publisher and school handling and cleanup procedures. Confirm the publisher alternate materials if no live kit.

Print per student

Investigate Mushrooms, Student Bundle PDF 103–106, one set per student (about 22 per class). Hand it out before the expert groups collect evidence. For the community activity, use PDF 139 or have students write directly in their notebooks.

Print per expert group

One copy of the relevant source instructions per expert group (Teacher Guide printed pp. 412–414). Use color if available, or show slides I–J for the corresponding images. Set copies out by source before D–J.

Teacher only

Keep setup and handling guidance (printed pp. 394, 410–411) and formative guidance (p. 396) handy. Save completed source notes for the mixed-group share.

Materials checklist

Use with Prep Desk p. 098 for exact quantities, copy ranges and safety notes.

Class or section: _____ Date: _____

- [ ] Start the live mushroom kit at least a week ahead if using it, and verify the image option if needed.
- [ ] Stage the three evidence sources, group labels, and hand lenses as directed.
- [ ] Copy Investigate Mushrooms for each student and prepare one source guide per expert group.
- [ ] Prepare the Mushroom Comparison chart and plan expert-to-mixed group changes and cleanup.

SET ASIDE FOR THE NEXT SECTION

THE SCIENCE BRIEFING

Mushrooms grow on a log. Where does their growth come from?

HOST SEGMENT / UNIT 5.1 / LESSON 12

WHERE WE ARE IN THE STORY

Termites gave us one way to think about getting matter from wood. Mushrooms raise a different question. Have groups bring different observations together before deciding what those observations can explain. Keep the comparisons handy for Lesson 13, when we'll trace what happens to matter during decomposition.

Here is the mystery. After investigating termite digestion, students compare evidence about mushrooms. Ask them to track what each source shows before comparing how fungi, plants and animals grow.

1. Recall where we left off | 10 min. Revisit what we figured out about microscopic organisms in our last lesson and recall that next we wanted to explore other organisms in the nurse log system to see if they also break down the nurse log.

2. Make observations | 30 min. Investigate one data source about mushrooms and then work with a mixed group to look for patterns across our findings and compare them to how plants and animals get matter and energy.

3. Participate in a discussion | 25 min. Share our findings in a Building Understandings Discussion and look for patterns in the evidence across groups, to help figure out where mushrooms get matter and energy to grow.

4. Connect to our communities | 20 min. Individually, we look for mushrooms and related living things in our own community and figure out that mushrooms are an example of fungi.

5. Decide where to go next | 5 min. Determine that we still have not answered our questions about if fungi can break down the nurse log, so we decide to start with this next time.

Where it leads. The class carries these findings into Lesson 13 to explain where matter goes during decomposition.

OPENSCIED UNIT 5.1 / TEACHER GUIDE PRINTED PP. 391–393

Stop and restart

STOP

If K–L cannot start cleanly, pause after D–J.

SAVE

Source notes and mixed-group comparison sheets.

RESTART

Begin the discussion. Flex the community activity or schedule an extra block to preserve the evidence comparison.

Track each section separately

Use the existing understanding record to mark the last activity and saved work. Treat the proposed break as flexible.

PICK UP THE LESSON

Prep Desk / p. 098

Teaching Moves / p. 102

Field Notes / p. 104

How do mushrooms get what they need to grow?

Science and storyline

Carry forward the termite evidence. Students compare sources to distinguish fungi from plants and animals. Dark growth is evidence against needing light for growth in this case; it does not by itself identify a complete source of matter or energy. Lesson 13 develops the decomposer explanation.

A–C / 10 MIN

Revisit fungi questions.

D–J / 30 MIN

Assign source, collect evidence and regroup to compare it.

Run it. Assign expert groups across the three sources. Record observations on Investigate Mushrooms, then regroup with one source represented by each expert. Compare only after students record peers' evidence.

Ask. Which source did you use? What did you observe?

Listen and respond. If a group lacks a source, have an expert or the teacher provide that source's recorded observation. Save source notes and comparison sheets at the bell.

Teacher Guide printed pages. 399–402

How do mushrooms get what they need to grow?

K–L / 25 MIN

Discuss patterns across sources.

Run it. Use the Mushroom Comparison chart to bring claims together. Press for evidence across sources and invite competing interpretations.

Ask. Which result supports that idea? What remains uncertain about matter or energy?

Listen and respond. If students say dark growth proves where matter comes from, separate the evidence about light from the question of matter source.

Teacher Guide printed pages. 403–406

Teacher cue. Ask for one observation from each of two sources. If students name a conclusion alone, ask which feature or comparison supports it.

M–O / 20 MIN

Connect to community examples.

Run it. Use community examples to apply the comparison. Ask students which idea is supported by their observations and which raises a new question.

Ask. How is your example like what we compared?

Listen and respond. Save open questions for decomposition. If time is short, preserve the evidence discussion and flex the community activity.

Teacher Guide printed pages. 406–408

P / 5 MIN

Decide what to investigate next.

Access during instruction

During the jigsaw, let a student point to the image or specimen and use a short oral report to name the source, observation, and claim.

Understanding record

One record per class section

CLASS / SECTION	DATE	ACTIVITY / SLIDE

Likely ideas and next moves
Mushrooms are plants because they stay still: ask what light/dark evidence adds. They eat like animals: ask what structures were observed and what cannot be inferred. Require source labels so students can check each claim.

Evidence checkpoint
Ask students to use one observation from their source and a second from a peer. If they repeat a claim without evidence, have them find the recorded feature, condition or comparison that supports it.

Evidence students used

Ideas or questions to revisit

Last completed activity and saved work

First move next time

SCIENCE STUDIO

WHERE DOES IT GO?

GRADE 5 / UNIT 5.1 / LESSON 13 / BILL NYE



Setup checklist

SET UP REQUIRED

DO BEFORE CLASS

• Queue the flower decomposition video and prepare the fungi article, evidence collector and class model.

PUBLISHER MATERIALS

Per student:

• Communities Celebrate National Mushroom Month newspaper article • Fungi Evidence Collector handout • Notice and Wonder: Decomposers and Matter • Model a Mushroom • My Growing Ideas

Per partners:

• Fungi Evidence Collector • Model a Mushroom

Per class:

• Flower Decomposition video • Nurse Log Consensus Model chart • previous versions of consensus model

Source: Teacher Guide printed p. 420 (PDF viewer p. 426).

Before class

G420. Prepare the fungi article for each student to use in reading pairs, or share the publisher slide deck. Queue the flower decomposition video, prepare the evidence collector and retrieve the class model. Student Bundle PDF 107, 109, 111,

113. Confirm pair and individual copy jobs.

Print and handout map

Student Bundle viewer pages unless marked Teacher Guide printed. Prepare one per student unless otherwise stated (about 22 per class).

Student copies

One Fungi Evidence Collector (107), Notice and Wonder: Decomposers and Matter (109), and Model a Mushroom (111) per student. Students use their own sheets during partner work. Copy My Growing Ideas (113) for the close or use notebooks.

Shared reading and teacher prep

Give each student access to the publisher mushroom-month article for partner reading, using individual copies or the publisher slide deck. Give the evidence collector at the reading, the Notice and Wonder page with slide H, and the model sheet at partner modeling. Queue the flower video and have earlier class models available.

Materials checklist

Use with Prep Desk p. 106 for exact quantities, copy ranges and safety notes.

Class or section: _____ Date: _____

- [ ] Prepare the mushroom-month article for each student's use during paired reading, or share the publisher slide deck.
- [ ] Queue the flower decomposition time-lapse and gather the earlier class models.
- [ ] Copy Fungi Evidence Collector, Notice and Wonder, and Model a Mushroom.
- [ ] Set out the comparison and modeling space. Prepare Growing Ideas or notebooks.

SET ASIDE FOR THE NEXT SECTION

THE SCIENCE BRIEFING

When something decays, where does its matter go?

HOST SEGMENT / UNIT 5.1 / LESSON 13

WHERE WE ARE IN THE STORY

Last time students compared evidence about mushroom growth. Now we're following matter as once-living things break down. Ask students to connect observations to their models, then use those ideas to revise the class nurse-log model. Next we'll test what happens when a new organism changes the web.

Here is the mystery. Lesson 12 raised questions about what fungi use to grow. Here students use reading, a time-lapse and models to explain the role of decomposers in the system.

1. Recall where we left off | 5 min. Recall the questions we had about where mushrooms and other fungi get their matter and energy from.

2. Read an article | 20 min. With a partner, we read an article about mushrooms and other fungi in different communities to collect evidence about where they get matter and energy to grow.

3. Participate in a discussion | 10 min. Discuss what we figured out about mushrooms and other fungi from reading the article, and categorize them as decomposers.

4. Make observations | 10 min. Watch a time-lapse video of a decaying flower that includes decomposers and consider what would happen if the system did not include decomposers.

5. Develop a model to explain | 15 min. In partners, we model to explain how decomposers like mushrooms get and use matter and energy.

6. Revise our model to explain | 20 min. Add new ideas that we have figured out about fungi, decomposers, and their important role as recyclers of matter and energy to our Nurse Log Consensus Model.

7. Update our Growing Ideas charts | 5 min. Individually update our Growing Ideas charts to document our new ideas about fungi, decomposers, and how matter and energy move through the nurse log system.

8. Generate new ideas | 5 min. With a partner, we discuss what we think happens when other organisms besides decomposers are removed from or added to ecosystems.

Where it leads. Once the class has modeled these relationships, Lesson 14 asks what changes when an organism enters the system.

OPENSCIED UNIT 5.1 / TEACHER GUIDE PRINTED PP. 417–419

Stop and restart

STOP

Finish G–J.

SAVE

Article evidence and time-lapse observations.

RESTART

Start K with a visible change students can explain, then introduce the model task.

Track each section separately

Use the existing understanding record to mark the last activity and saved work. Treat the proposed break as flexible.

PICK UP THE LESSON

Prep Desk / p. 106

Teaching Moves / p. 110

Field Notes / p. 112

Where does matter go when things decompose?

Science and storyline

Lesson 12 compared fungi with plants and animals. Use the article and time-lapse to model decomposition as matter moving and changing form. The time-lapse alone cannot show all destinations of matter.

A–B / 5 MIN

Reopen fungi questions.

C–D / 20 MIN

Read and collect evidence.

Run it. Give each student the Fungi Evidence Collector and access to the mushroom-month article. Have partners read and each record what the source says fungi did.

Ask. What does the article say changed? Which detail supports it?

Listen and respond. If partners give only a name, ask what the fungus did and where the passage explains it.

Teacher Guide printed pages. 425–428

E–F / 10 MIN

Discuss the decomposer role.

G–J / 10 MIN

Observe the time-lapse.

Run it. Collect visible changes from the flower time-lapse before asking where matter went. Revisit a useful moment as directed.

Ask. What changed that we could see? What can't the video track?

Listen and respond. If students say matter is gone, record it as an idea to evaluate. Save observations alongside article evidence.

Teacher Guide printed pages. 429–432

Where does matter go when things decompose?

K / 15 MIN

Model the change.

Teacher cue. Ask students to label where the matter starts and where it goes. Have partners connect an arrow to the article or time-lapse evidence.

L–N / 20 MIN

Revise the class model.

Run it. Compare Model a Mushroom explanations and revise the class model. Trace a pathway students can support.

Ask. What is being broken down? Where might some of its matter go?

Listen and respond. If a model ends at disappearance, have students draw a possible destination and label the inference.

Teacher Guide printed pages. 435–440

O / 5 MIN

Update Growing Ideas.

P / 5 MIN

Generate questions.

Access during instruction

During partner modeling, offer arrows and labels or oral rehearsal to show where matter starts and where it goes.

Understanding record

One record per class section

CLASS / SECTION	DATE	ACTIVITY / SLIDE

Likely ideas and next moves
It vanished: ask what changed visibly and where matter might have moved. The model uses one arrow for everything: ask the student to label the matter movement separately from energy transfer.

Evidence checkpoint
Ask students to annotate one model arrow and point to article or video evidence. If they infer an unobserved destination, mark that as an inference and ask which source supports it.

Evidence students used

Ideas or questions to revisit

Last completed activity and saved work

First move next time

SCIENCE STUDIO

BULLFROG INTERRUPTION

GRADE 5 / UNIT 5.1 / LESSON 14 / MR. WIZARD



Setup checklist

SET UP REQUIRED

DO BEFORE CLASS

• Prepare the game cards, two yarn colors and clips. Set aside the bullfrog card until the publisher calls for its introduction.

PUBLISHER MATERIALS

Per student:

• 2-3 clothespins or paper clips • Food Web Exit Slip • My Growing Ideas

Per small group:

• 6 Facts About Bullfrogs • American bullfrog game card from the Energy and Matter Flow Game Cards reference

Per class:

• Class set of the Energy and Matter Flow Game Cards reference • components of the nurse log system chart • ball of yellow yarn • ball of yarn of any other color

Source: Teacher Guide printed p. 447 (PDF viewer p. 453).

Before class

G447 and Game Guidance. Prepare one game card per student, 2–3 clips each, two yarn colors, components chart, infographic and bullfrog card. Keep the introduction card aside until instructed. Plan space and reset for both sections; keep exit sheets separate. Student Bundle PDF 115–117.

Print and handout map

Student Bundle viewer pages unless marked Teacher Guide printed. Prepare one per student unless otherwise stated (about 22 per class).

Game copies

Prepare one role card per student from the publisher game set. English base cards are in guide printed pp. 488–502, extra roles pp. 507 onward. Set aside the bullfrog cards (pp. 503–506). Prepare one 6 Facts About Bullfrogs infographic and bullfrog card per group for K–L. Introduce the role in Round 2.

Student and teacher copies

One Food Web Exit Slip (115–116) per student for individual work at the end. Use My Growing Ideas (117) for P or continue in notebooks. Keep Game Guidance (printed pp. 463–465) at the teacher station.

Materials checklist

Use with Prep Desk p. 114 for exact quantities, copy ranges and safety notes.

Class or section: _____ Date: _____

- [ ] Set out one role card and two or three clips per student, plus two colors of yarn.
- [ ] Set aside the bullfrog cards and infographic for the publisher’s later step.
- [ ] Prepare the components chart and individual Food Web Exit Slips.
- [ ] Plan how to capture the baseline web and reset the game for the next section.

SET ASIDE FOR THE NEXT SECTION

THE SCIENCE BRIEFING

A bullfrog enters the food web. What changes?

HOST SEGMENT / UNIT 5.1 / LESSON 14

WHERE WE ARE IN THE STORY

Our class model now connects producers, consumers, and decomposers. The game gives students a way to see those connections in action. Compare the web before and after the bullfrog enters. Keep that reasoning available as students move to a new ecosystem in the final lesson.

Here is the mystery. Students have traced matter and energy through parts of the nurse-log system. The game lets them represent those connections and think through what happens when the web changes.

1. Recall where we left off | 5 min. Recall what we figured out about the nurse log system in our last lesson and start thinking about which component of the nurse log system is the most important.

2. Play and investigate | 35 min. Each take on the role of one of the components of the nurse log system while playing a game that helps us visualize a balanced food web.

3. Read an infographic | 15 min. Read an infographic about American bullfrogs.

4. Play and investigate | 20 min. Play the Matter and Energy Flow game with the addition of the American bullfrog.

5. Reflect on our work | 10 min. Individually, we use an exit slip to think through our ideas about how the American bullfrog impacts the nurse log food web.

6. Revisit our Growing Ideas charts | 5 min. Individually, we add what we figured out about how the balance and health of a food web can be disrupted to our Growing Ideas charts.

Where it leads. Save the bullfrog predictions and evidence. Lesson 15 takes the food-web ideas into a different ecosystem.

OPENSCIED UNIT 5.1 / TEACHER GUIDE PRINTED PP. 444–446

Stop and restart

STOP

Before introducing the bullfrog, pause if needed.

SAVE

A sketch or photo of the baseline web and the infographic prediction if started.

RESTART

Resume K–L before introducing the bullfrog. Reconstruct only the connections needed for comparison.

Track each section separately

Use the existing understanding record to mark the last activity and saved work. Treat the proposed break as flexible.

PICK UP THE LESSON

Prep Desk / p. 114

Teaching Moves / p. 118

Field Notes / p. 120

What changes when a bullfrog enters the web?

Science and storyline

The class has traced matter and energy through the nurse-log system. The game makes connections visible and supports reasoning about a disruption. Game changes represent possible effects and need interpretation.

A–B / 5 MIN

Revisit the system.

C–J / 35 MIN

Play and document the baseline web.

Run it. Assign baseline roles and use the yarn colors according to Game Guidance. Capture the starting web before adding a species.

Ask. What does this strand represent? Which two parts does it connect?

Listen and respond. If students reverse the relationship, name source and receiver and trace it. Save a photo or sketch before resetting.

Teacher Guide printed pages. 451–457

K–L / 15 MIN

Read the infographic and predict.

Run it. Give groups the infographic and bullfrog card. Have them use evidence to complete the card and predict a change before Round 2.

Ask. Which fact supports your prediction?

Listen and respond. If the prediction is vague, ask students to name one connection and the organism affected. Keep the card aside until Round 2.

Teacher Guide printed pages. 458

What changes when a bullfrog enters the web?

M–N / 20 MIN

Add the bullfrog and compare.

Run it. Introduce the bullfrog during the second round. Compare one changed pathway to the baseline before the individual exit slip at O.

Ask. Which connection changed first? What else might be affected, and why?

Listen and respond. Trace a direct effect before an indirect effect. Leave the independent exit-slip response to each student.

Teacher Guide printed pages. 459–460

Teacher cue. Compare the two rounds. Ask students to trace one direct effect of the bullfrog before proposing an indirect effect.

O / 10 MIN

Reflect using evidence.

P / 5 MIN

Update Growing Ideas.

Access during instruction

After the game and before the independent exit slip, let students trace one yarn connection and explain a direct effect before discussing indirect effects.

Understanding record

One record per class section

CLASS / SECTION	DATE	ACTIVITY / SLIDE

Likely ideas and next moves
Only the organism eaten is affected: trace one direct change and then follow a second connection. Everything changes the same way: compare the baseline with the altered model and ask which relationships were actually represented.

Evidence checkpoint
Use the exit response to check for a direct or indirect effect supported by a connection. If the response names only an organism, ask the student to describe the link and what changed.

Evidence students used

Ideas or questions to revisit

Last completed activity and saved work

First move next time

SCIENCE STUDIO

GRASSLAND CONNECTIONS

GRADE 5 / UNIT 5.1 / LESSON 15 / BILL NYE AND MR. WIZARD



UNIT SIGNAL / 15 OF 15

Setup checklist

SET UP REQUIRED

DO BEFORE CLASS

• Prepare the grassland organism cards by pair. Queue Ecosystem Explorers and separate the student assessment from scoring materials.

PUBLISHER MATERIALS

Per student:

• Grassland Ecosystem Food Chains • Grassland Ecosystem

Per partners:

• Grassland Ecosystem Organism Cards

Per small group:

• Class Consensus Model

Per class:

• Ecosystem Explorers book

Source: Teacher Guide printed p. 513 (PDF viewer p. 519).

Before class

G513, 515 and 529 onward; S119 food chains, S121–125 organism cards, S151–155 assessment. Print one card set per pair (about 11/class), cut and fold as directed; prepare bullfrog cutouts and tape, markers, prior class model and Ecosystem Explorers. Print the food chains and assessment per student as assigned. Keep scoring guidance teacher-only. Check the guide and school conditions before giving any supports during assessment.

Print practice materials

One Grassland Ecosystem Organism Cards set per pair (Student Bundle PDF 121–125, about 11 sets per class). Cut on solid lines and fold on dotted lines before D–G. Print one Food Chains handout per student (PDF 119) for H–I. Prepare the bullfrog cutouts from Teacher Guide printed p. 527.

Hold for slide J

One Grassland Ecosystem assessment per student (Student Bundle PDF 151–155, about 22 sets per class). Keep it separate from practice. Read the scenario and take task questions as directed. Reserve 30 minutes of independent work per printed p. 525. The snapshot lists 25 minutes for J, so schedule extra time if needed.

Teacher only

Keep the scoring guide (printed pp. 529–531) with teacher materials. Use it after collecting student work. Confirm school-required accommodations and the publisher directions before administration.

Materials checklist

Use with Prep Desk p. 122 for exact quantities, copy ranges and safety notes.

Class or section: _____ Date: _____

- [ ] Prepare grassland organism card sets by pair and cut or fold them as directed.
- [ ] Bring the class consensus model, Ecosystem Explorers, markers, and the bullfrog cutouts.
- [ ] Copy Grassland Ecosystem Food Chains for practice.
- [ ] Set aside the individual assessment for slide J and keep scoring guidance teacher-only. Reserve the full independent work window.

SET ASIDE FOR THE NEXT SECTION

THE SCIENCE BRIEFING

Can the nurse-log ideas explain a grassland?

HOST SEGMENT / UNIT 5.1 / LESSON 15

WHERE WE ARE IN THE STORY

We've used the nurse log to trace matter and energy and to reason about a disruption. Now students get a chance to use those same ideas in a grassland. Let the food-chain practice prepare them for the independent task, then use their work to see what transfers to the new system.

Here is the mystery. Students know how organisms in the nurse-log system are connected. Today they transfer those ideas to a grassland and explain how matter and energy move through its food web.

1. Recall where we left off | 5 min. Look back on what we figured out from the Matter and Energy Flow game.

2. Revise our Class Consensus Model to explain | 10 min. Add the ideas we figured out about American bullfrogs being added to the nurse log system to our Class Consensus Model.

3. Read a book | 10 min. Read Ecosystems Explorers and discuss the different ecosystems we observed.

4. Identify related phenomena | 15 min. In pairs, we predict and discuss where organisms in a grassland ecosystem get and give matter and energy.

5. Make and compare predictions | 20 min. Create grassland food chains and communicate our food chains to the class.

6. Apply ideas to another phenomenon | 25 min. Individually, we develop a model of what happens when a wild hog is introduced to a grassland ecosystem.

7. Consider what is next | 5 min. Celebrate our learning and look forward to learning more.

Where it leads. Use the publisher's transfer task to see how students apply the ideas in a new setting. Keep their work separate from teacher scoring guidance.

OPENSCIED UNIT 5.1 / TEACHER GUIDE PRINTED PP. 510–512

Stop and restart

STOP

If practice runs long, schedule a separate assessment window.

SAVE

Card arrangement or sketch and the last completed H–I step.

RESTART

Protect the full J window next block, including 30 minutes of independent work.

Track each section separately

Use the existing understanding record to mark the last activity and saved work. Treat the proposed break as flexible.

PICK UP THE LESSON

Prep Desk / p. 122

Teaching Moves / p. 126

Field Notes / p. 128

Can the nurse-log model explain a grassland?

Science and storyline

Use Lesson 14's bullfrog disruption to bridge to a new ecosystem. Matter moves among organisms and the environment; energy enters through producers and transfers through feeding. Keep class practice and the individual wild-hog transfer task separate.

Assessment timing

The snapshot lists 25 minutes for J. The detailed directions (printed p. 525) allow 30 minutes of independent work. Plan the class block around that full window.

A / 5 MIN

Revisit the game.

B / 10 MIN

Revise the nurse-log consensus model.

C / 10 MIN

Read the ecosystems book.

D–G / 15 MIN

Predict using the organism cards.

Run it. Introduce the grassland setting. Have partners inspect the organism cards before arranging relationships.

Ask. What does this card say the organism gets or gives?

Listen and respond. If students assume a nurse-log relationship carries over, send them to the grassland card for evidence.

Teacher Guide printed pages. 521–522

Can the nurse-log model explain a grassland?

H–I / 20 MIN

Make and compare grassland food chains.

Run it. Use the Food Chains handout for practice. Compare arrangements and have students explain arrows before the separate transfer task.

Ask. Which organism is the source, which is the receiver, and what moves?

Listen and respond. If an arrow is reversed, have students read the relationship aloud using the cards. Save a sketch and the last finished step if pausing.

Teacher Guide printed pages. 523–524

J / ASSESSMENT

Administer the transfer task under publisher conditions.

Run it. Distribute the assessment at J, read the scenario, and take task questions. Keep the 30-minute independent work window continuous. Follow the guide's access options and required accommodations.

Ask. Clarify directions as permitted. Keep practice examples and scoring notes separate.

Listen and respond. Use the scoring guide after collection to plan follow-up on transfer. Keep instructional feedback for after the task.

Teacher Guide printed pages. 525

Teacher cue. Read the scenario and take task questions. Reserve 30 minutes for independent work. Use the publisher's access guidance on printed p. 525 for response options and references.

K / 5 MIN

Close.

Access during instruction

During the food-chain practice before slide J, have students point to the source and receiver for one arrow and explain what moves. Keep this rehearsal separate from the independent task.

Understanding record

One record per class section

CLASS / SECTION	DATE	ACTIVITY / SLIDE

Likely ideas and next moves
During practice, if an arrow is reversed, ask what moves from which organism to which. If students say the whole web will change immediately, ask them to trace one direct effect before an indirect one. Keep prompts about the actual assessment scenario for scoring and follow-up.

Evidence checkpoint
Before J, check one chain for arrow direction and whether students can describe matter or energy transfer. During J, preserve independent work. Use the official rubric, record what the model shows, and follow the guide for individual clarification when needed.

Evidence students used

Ideas or questions to revisit

Last completed activity and saved work

First move next time

Teaching notes

Use for a lesson, prep rehearsal, or follow-up.

CLASS / SECTION _____ **DATE** _____ **LESSON / TOPIC** _____

NOTES

FOLLOW-UP / WHAT TO HAVE READY NEXT TIME